MATHEMATICS GRADE 10: REAL NUMBERS

CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 1.1 (7 Lessons)

SUB-STRAND OVERVIEW	
Grade Level	10
Subject	Mathematics
Strand	Strand 1.0: Numbers and Algebra
Sub-Strand	Sub-Strand 1.1: Real Numbers
Total Duration	7 lessons × 40 minutes = 280 minutes total
Sub-Strand Content	– Classification of whole numbers as odd, even, prime, and composite – Classification of real numbers as rational and irrational numbers – Reciprocal of real numbers by division – Reciprocal of real numbers using mathematical tables – Reciprocal of real numbers using calculators and digital devices – Application of reciprocals in mathematical computations – Real numbers in day-to-day activities
Learning Outcomes	By the end of the sub-strand, the learner should be able to: a) classify whole numbers as odd, even, prime and composite in different situations, b) classify real numbers as rational and irrational in different situations, c) determine the reciprocal of real numbers by division, d) determine the reciprocal of real numbers by use of mathematical tables and calculators, e) apply reciprocal of real numbers in mathematical computations, f) promote the use of real numbers in day-to-day activities.
Core Competencies	– Communication and Collaboration: the learner discusses with peers to identify and categorise whole numbers as odd, even, prime, composite, rational, and irrational. – Critical Thinking and Problem Solving: the learner uses the reciprocal of real numbers to work out mathematical tasks. – Digital Literacy: the learner works out the reciprocal of real numbers from calculators and other digital devices.
Core Values	– Unity: the learner discusses in a group and categorises real numbers as rational and irrational numbers. – Responsibility: the learner takes care of calculators and mathematical tables when working out the reciprocal of real numbers.
Science & Engineering Practices	– Asking Questions and Defining Problems: learners pose questions about why some numbers (e.g. $\sqrt{2}$, π) cannot be expressed as exact fractions, driving inquiry into irrational numbers. – Developing and Using Models: learners use pizza-slice diagrams and fraction circles as visual models to represent rational numbers and their reciprocals. – Using Mathematics and Computational Thinking: learners compute reciprocals by division, via mathematical tables, and with

	calculators, comparing methods for accuracy and efficiency. – Obtaining, Evaluating, and Communicating Information: learners read reciprocal values from mathematical tables, evaluate results against calculator outputs, and communicate findings to peers.
Pertinent & Contemporary Issues (PCIs)	– Life Skills (Self-Esteem): the learner participates in the discussion on the concepts of rational and irrational numbers, building confidence in mathematical reasoning.
Career Connections	– Engineering and Construction: understanding real numbers and reciprocals is essential for calculations in structural design and quantity surveying, e.g. scaling materials for buildings along Nairobi's Thika Superhighway. – Agriculture and Food Science: farmers in the Rift Valley use rational number calculations for mixing fertiliser ratios and dividing land parcels; reciprocals help in rate and proportion problems. – Finance and Banking: bank clerks and M-Pesa agents apply number classification and reciprocal computations when converting interest rates and splitting shared costs. – Healthcare and Pharmacy: nurses at Kenyatta National Hospital use reciprocals when calculating drug dosage rates and dilution ratios. – Data Science and Technology: software developers and data analysts classify and process numerical data types — a foundation rooted in understanding real number categories.
Focus for Lessons	This unit introduces Grade 10 learners to the full landscape of real numbers — from the familiar categories of odd, even, prime, and composite whole numbers to the broader distinction between rational and irrational numbers. Using Kenya-relevant contexts such as sharing ugali portions, dividing maize harvest yields in Eldoret, and reading mathematical tables common in Kenyan secondary schools, learners build conceptual clarity about how numbers are structured. A central thread throughout the unit is the concept of the reciprocal, explored through manual division, mathematical tables, and digital calculators, equipping learners with multiple computational tools for real-world problem solving.
Driving Question / Key Inquiry Questions	DRIVING QUESTION: Why can some numbers be written as exact fractions and others cannot — and how does knowing a number's "type" help us solve everyday problems in Kenya? KICD KEY INQUIRY QUESTION: 1. How do we use real numbers in day-to-day activities?
Anchoring Phenomenon	ANCHORING PHENOMENON — The Unequal Pizza Slices Problem: A group of friends in Nairobi order a pizza and attempt to share it equally. When they try to describe each person's share as a fraction, they notice that some cuts produce neat, expressible fractions (e.g. 1/4, 1/8), while other cuts — like trying to divide the pizza into 3 equal parts and measuring the arc length — produce decimals that never end and never repeat (approximately 0.3333… vs. $\sqrt{2} \div 2 \approx 0.7071\ldots$). Students observe two pizza diagrams: one divided into equal rational-fraction slices, and one where a slice size is expressed as an irrational number. This is puzzling because both slices look like ordinary pieces of pizza — they exist physically — yet one can be written as a neat fraction and the other cannot. The puzzle deepens when students ask: if we know the whole pizza and the number of slices, can we always find how many slices fit into the whole? This is the concept of the reciprocal — and students discover it works differently for rational and irrational numbers. Fraction circles are used as a hands-on manipulative to anchor this exploration throughout the unit.
Storyline Thread	Lesson 1 — What Kind of Number Is It? (Whole Number Classification): Students examine the pizza diagram and fraction circles. They sort a set of numbers (2, 3, 4, 15, 17, 36…) into odd, even, prime, and composite categories using peer discussion. The anchoring phenomenon is introduced: why do some pizza slices produce neat numbers and others do not? Students record initial predictions on the Driving Question Board (DQB).

	Lesson 2 — Rational vs. Irrational: The Number Line Story: Students place numbers on a number line and attempt to express √2, π, and 0.333… as fractions. Through guided group discussion, they categorise real numbers as rational or irrational and connect back to the pizza diagram — which slices are "rational" and which are "irrational"? The DQB is updated with new vocabulary. Lesson 3 — Flipping Numbers: Introducing Reciprocals by Division: Using the maize-bag distribution scenario from Eldoret, students discover that dividing 1 by a number gives its reciprocal. They work out reciprocals of whole numbers and simple fractions by hand, connecting the process to the pizza model — "how many slices fit into one whole?" Fraction circles are used as manipulatives. Lesson 4 — Reading the Tables: Reciprocals from Mathematical Tables: Students are issued mathematical tables (as used in Kenyan national exams). They practise locating reciprocal values for 2-, 3-, and 4-digit numbers, cross-checking with manual division results from Lesson 3. Teacher models precise table-reading technique; students work in pairs and record discrepancies, building responsibility for careful tool use. Lesson 5 — Digital Tools: Reciprocals on Calculators and Devices: Students use scientific calculators and, where available, digital devices to compute reciprocals of decimals, fractions, and irrational approximations (e.g. 1/√2). They compare calculator results with table values, discussing sources of rounding differences. The M-Pesa bill-splitting supporting phenomenon is revisited to apply reciprocals in a financial context. Lesson 6 — Applying Reciprocals: Solving Real-World Computation Tasks: Students tackle a structured set of problems drawn from Kenyan contexts — dividing fertiliser quantities, splitting market costs, and interpreting tile-cutting dimensions. They apply reciprocals from all three methods (division, tables, calculators) and justify which method is most appropriate for a given situation. The DQB is updated as more questions are answered. Lesson 7 — Model Building and Unit Synthesis: The Real Number Map: Students construct a cumulative "Real Number Map" — a visual classification diagram showing the hierarchy from natural numbers through integers, rationals, irrationals, to the full set of real numbers. Each category is annotated with a Kenyan real-world example. Students revisit the original pizza phenomenon and write a full explanation of why some slices are rational and others irrational, and how reciprocals apply to both. Final DQB review and peer presentation of models.
Supporting Phenomena	– Maize Bag Division in Eldoret: A farmer has 50 kg of maize to distribute equally among a number of families. When the number of families changes (3, 4, 5, 7…), some distributions yield neat decimal answers and others do not — connecting to rational vs. irrational classification and the use of reciprocals in rate calculations. – M-Pesa Transaction Splits: A group of Nairobi university students split a KSh 1,500 dinner bill. Students observe that splitting by 2, 4, or 5 yields whole-number shares, while splitting by 3 yields a repeating decimal (KSh 500.000… vs. KSh 499.999…), illustrating the concept of rational numbers in financial contexts. – Prime Number Security (Mpango wa Akili): A cybersecurity class in Mombasa learns that online banking PINs and encryption rely on the properties of prime numbers — making the classification of primes and composites directly relevant to digital safety in Kenya. – Tile Cutting on a Construction Site: A fundi (artisan) in Kisumu cuts square tiles whose side length is √2 metres. When asked to express the exact length as a fraction, they cannot — providing a concrete, local construction context for irrational numbers.

LESSON 1 (80 minutes): What Kind of Number Is It? — Launching the Pizza Phenomenon	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To launch the anchoring phenomenon and introduce the classification of whole numbers as odd, even, prime, and composite, connecting number type to real-world sharing problems in Kenya.
Knowledge	Learners will know that whole numbers can be classified as odd, even, prime, or composite, and that this classification is the first step toward understanding why some numbers produce 'neat' fractions and others do not.
Skills	Learners will be able to sort and justify the classification of whole numbers as odd, even, prime, or composite using examples drawn from the pizza-sharing phenomenon.
Attitudes	Learners will appreciate (Unity) the value of discussing and categorising numbers collaboratively in a group, and will demonstrate (Responsibility) careful use of fraction-circle manipulatives and shared materials.
Key Inquiry Question	How do we use real numbers in day-to-day activities?
Purpose in Storyline	This is the opening lesson of the unit. It launches the anchoring phenomenon — the Unequal Pizza Slices Problem — and opens the Driving Question Board. Students begin to notice that the NUMBER of slices chosen determines whether a share is expressible as a neat fraction. This puzzles them enough to ask: what makes some numbers 'nicer' than others? That question drives the entire unit.
Safety Notes	No chemical or physical hazards. Remind learners to handle printed fraction-circle manipulatives and scissors (if cutting) carefully. Ensure equitable group participation — monitor for dominant voices and ensure quieter learners are cold-called.

B. LESSON OVERVIEW

Students in Nairobi are introduced to a real-world puzzle: a group of friends orders a pizza and tries to share it equally. Two pizza diagrams are presented — one divided into 4 equal slices (each share = 1/4, a neat fraction) and one divided into 3 equal slices where the arc length of each slice involves a decimal that never ends and never repeats. Students use fraction-circle manipulatives to physically explore different ways of dividing a 'pizza'. They then sort numbers of slices (1 through 12) into odd/even and prime/composite categories, noticing patterns in which divisions feel 'clean'. The lesson closes by opening the Driving Question Board with student-generated questions anchored to the phenomenon. No formal instruction on rational/irrational numbers is given yet — this lesson is entirely about noticing, predicting, and questioning.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** [Converting a fraction to a repeating decimal] Source: Khan	Each learner receives a printed fraction-circle worksheet showing a blank circle labelled 'Pizza — 1 Whole'. The teacher poses the scenario: 'A group of friends in Nairobi order one pizza from a	Distribute fraction-circle worksheets before speaking. Read the scenario aloud twice, slowly. Say exactly: 'I want SILENT thinking for 4 minutes — no talking, no asking me questions.	PREDICT-BEFORE-INSTRUCTION (Elicitation of Prior Knowledge): By asking students to predict before any classification vocabulary is introduced, the teacher surfaces intuitive ideas	OBSERVABLE INDICATOR: Can the learner write a valid fraction (e.g. 1/4, 1/6) to represent one person's share for at least two of their chosen numbers? Circulate and look for: (a) students who

Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Real Number Line Graphs (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	restaurant along Kimathi Street. They want to share it EXACTLY equally — no one gets more, no one gets less.' Students are asked to silently write their prediction in their exercise books: (1) List THREE different numbers of friends that would make sharing feel 'clean and easy'. (2) List TWO numbers of friends that you think would make sharing feel 'messy or impossible to describe exactly'. (3) For each number you chose, write what fraction of the pizza ONE person would get. Students work individually and silently for 4 minutes before any discussion.	Write your predictions in your exercise book. There is no wrong answer right now. Go.' Set a visible 4-minute timer. During wait time, circulate and make brief tally marks on a notepad noting which numbers students write most frequently — use this data to sequence the class discussion. After 4 minutes, say: 'Turn to your neighbour — you have 90 seconds. Share your two lists. Do you agree or disagree on which numbers feel clean? Why?' After 90 seconds, cold-call FOUR students (pick two who wrote 'clean' numbers and two who wrote 'messy' numbers from your tally). Ask each: 'Tell us one number you predicted feels clean — and explain WHY you think so in one sentence.' Apply 10-second wait time before accepting any answer.	about divisibility and fractions. This reveals prior knowledge gaps and misconceptions (e.g. students who think 6 is 'messy' or that 7 is 'easy') that will be explicitly addressed in Phase 3. Writing predictions first also gives every learner — including quieter ones — a committed position to defend or revise.	write decimals instead of fractions — note them for targeted questioning in Phase 3; (b) students who cannot write any fraction — pair them with a peer during the turn-and-talk. Collect no books — this is a low-stakes thinking record.
Observe Phase 📹 **VIDEO:** Converting a fraction to a repeating decimal Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Real Number Line Graphs (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	The teacher displays (or distributes printed copies of) two pizza diagrams side by side on the board: – PIZZA A: Divided into 4 equal slices. Each slice is labelled '1/4'. The decimal equivalent '0.25' is shown — it terminates. – PIZZA B: Divided into 3 equal slices. Each slice is labelled '1/3'. The decimal equivalent '0.3333…' is shown with a bar over the 3, and the note: 'This decimal NEVER ends and NEVER stops repeating.' A third image is then revealed — PIZZA C — where the teacher says: 'Now imagine we are not counting slices, but measuring the straight-line distance (chord) across one slice of a pizza with a 28 cm diameter cut into 4 equal	Before revealing Pizza B, pause after Pizza A and ask the whole class: 'Thumbs up if 0.25 feels like a number you could work with easily when buying tomatoes at Wakulima Market.' (Expect nearly all thumbs up.) Then reveal Pizza B and the 0.3333… notation. Ask: 'Thumbs up if 0.3333… feels equally easy to work with.' (Expect hesitation.) Do NOT explain yet. Say: 'Write ONE question this image makes you want to ask — any question at all — in your book. You have 60 seconds.' After revealing Pizza C and √2 ÷ 2, cold-call THREE students to read their questions aloud without comment — say: 'I am not answering these yet. I am COLLECTING them.' Write each student question verbatim on a sticky note for the DQB in Phase 4. During the fraction-circle table activity (15	STRUCTURED OBSERVATION TABLE (Data Collection Before Explanation): Students gather their own evidence about decimal behaviour BEFORE the teacher names or explains rational/irrational numbers. This mirrors the NGSS Practice of Analysing and Interpreting Data. The physical act of folding fraction circles creates a tactile memory anchor — students can return to 'thirds feel different from quarters' throughout the unit. The three-pizza visual (A, B, C) is a concrete anchor for the three levels of complexity the unit will address.	OBSERVABLE INDICATOR: Can the learner correctly complete at least 5 of the 7 rows in the observation table, including correctly identifying 1/3 as a repeating decimal and 1/2 as a terminating decimal? Listen for: students who say 0.3333 'ends at some point' — this is a key misconception to address in Phase 3. Students who correctly write the bar notation ($0.\overline{3}$) demonstrate readiness for the rational number lesson in Lesson 3.

	parts. That distance involves √2. When we divide √2 by 2 we get approximately 0.7071067811…' This decimal is written on the board and students are asked to stare at it for 30 seconds. Students then use their fraction-circle manipulatives: they physically fold a paper fraction circle into halves, quarters, thirds, sixths, and fifths — recording in a table: Number of Slices | Fraction per slice | Decimal equivalent | Does the decimal terminate, repeat, or go on forever? Students complete the table for slices: 2, 3, 4, 5, 6, 8, 10.	minutes), circulate and ask probing questions to specific groups: 'You wrote that 1/5 = 0.2 — does that decimal end?' and 'You wrote 1/3 = 0.33 — is that the same as 0.3333… going on forever?' Apply 10-second wait time after each probe before moving on.		
Explain Phase 📹 **VIDEO:** Converting a fraction to a repeating decimal Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Real Number Line Graphs (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	The teacher now shifts focus from the fractions to the WHOLE NUMBERS used to count slices: 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12. Students work in groups of 4. Each group receives a set of 12 number cards (printed, one number per card) and a sorting mat with four labelled regions: ODD | EVEN | PRIME | COMPOSITE. Task 1 (8 minutes): Sort all 12 cards into ODD and EVEN. Groups must write one rule that explains how they decided. Task 2 (8 minutes): Now re-sort the same cards into PRIME and COMPOSITE. Groups must write: (a) the definition of a prime number they agreed on, and (b) identify any number that does NOT fit neatly into prime or composite (the number 1 — this surfaces the special case). Task 3 — Connection to Phenomenon (5 minutes): Groups answer: 'Look at your PRIME numbers. If you cut a pizza into a prime number of slices, can you re-cut it evenly into a smaller whole number of pieces without leaving leftover fractions? Explain using the pizza.' Groups record answers on a mini-	Introduce the sorting task with these exact words: 'You are NOT using the internet or your notes yet. Use only what you already know and what your group knows together. This is a COMMUNICATION AND COLLABORATION task.' While groups work on Task 1, circulate and listen. After Task 1, cold-call TWO groups to share their rule — ask the class: 'Does every group agree with that rule? Show thumbs up, thumbs sideways, or thumbs down.' For ANY thumbs sideways or down, ask that student: 'What would you change about the rule?' (10-second wait time.) For Task 2, anticipate the number 1 problem — when a group raises it, do NOT resolve it immediately. Say: 'Write that as a question for our board — that is an EXCELLENT puzzle. We will return to it.' For Task 3, ask each group to nominate a spokesperson DIFFERENT from the person who spoke last time. After all groups share Task 3, make this explicit transition statement: 'You have just discovered something important —	CARD SORT WITH JUSTIFICATION (Concept Categorisation): The physical sorting of number cards activates classification reasoning before formal definitions are given. Requiring groups to write their OWN rule for odd/even and their OWN definition of prime forces learners to construct understanding rather than receive it. Connecting prime numbers back to the pizza (Task 3) ensures the classification work is not abstract — it immediately serves the phenomenon. The deliberate surfacing of '1 as a special case' plants a productive puzzle that motivates deeper inquiry.	OBSERVABLE INDICATOR: Can the learner correctly classify all numbers 1–12 as odd/even AND identify at least 4 prime numbers from the set with a written justification? Listen for groups defining 'prime' as 'only divided by 1 and itself' — correct. Flag groups who include 1 as prime — address this explicitly in the next lesson. Check Task 3 responses: groups should recognise that a prime number of slices cannot be evenly re-divided — this is the intuitive seed of the concept that prime denominators produce non-terminating decimals (to be formalised in Lesson 3).

	whiteboard or large paper to share.	the TYPE of number we use to count slices affects whether the pizza shares feel clean or messy. THAT is the mystery we are going to spend the next 8 lessons solving.'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Converting a fraction to a repeating decimal Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Real Number Line Graphs (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	The teacher introduces the Driving Question Board — a large section of wall or board at the front of the classroom that will remain up for all 8 lessons of the sub-strand. The board has three columns labelled: WHAT WE NOTICE | WHAT WE WONDER | WHAT WE THINK WE KNOW. Students are given three sticky notes each (or write on paper that is taped up). They write: (1) ONE thing they noticed during today's lesson about the pizza or the numbers (NOTICE column). (2) ONE question they genuinely still have — something the lesson has not answered (WONDER column). (3) ONE thing they think they already know about real numbers or fractions (THINK WE KNOW column). Students walk up and post their notes. The teacher reads 5–6 notes aloud, grouping similar questions together with a bracket and a label (e.g. 'Three people are asking about why 1/3 never ends — let's call this the REPEATING DECIMAL CLUSTER').	Set up the DQB before learners post, with column labels clearly visible. As students write their notes (3 minutes, silent), circulate and identify 2–3 WONDER notes that directly connect to the unit's driving question ('Why can some numbers be written as exact fractions and others cannot?') — these will be read aloud first to anchor the board. After posting, read selected notes and say: 'This board belongs to YOU. Every lesson, we will come back here. Some of your questions will get answered. New ones will appear. By Lesson 8, we want this board to tell the story of what we learned.' Point to the driving question written at the top of the board: 'Why can some numbers be written as exact fractions and others cannot — and how does knowing a number's type help us solve everyday problems in Kenya?' Say: 'This is our compass question. Every lesson connects back to it.' Do NOT answer any questions posted on the board today — this is an opening, not a closing.	DRIVING QUESTION BOARD — NOTICE/WONDER/KNOW PROTOCOL: The DQB is a public, cumulative sensemaking artefact. Opening it in Lesson 1 with student-generated questions ensures learner ownership of the inquiry. The three-column structure (Notice / Wonder / Know) differentiates between empirical observation, genuine curiosity, and prior knowledge — helping students and teacher track conceptual change across the unit. Grouping similar questions into clusters models scientific discourse practices and shows students that their individual thinking contributes to collective understanding.	OBSERVABLE INDICATOR: Can each learner produce at least one substantive WONDER note that goes beyond surface observations (e.g. not just 'why is pizza round' but 'why does dividing by 3 give a decimal that never stops')? The teacher reads all WONDER notes after class and uses them to inform the sequencing emphasis of Lessons 2–4. Count the proportion of students whose WONDER notes reference the decimal behaviour or the number classification — this indicates readiness for Lesson 2.
Model Building Phase 📹 **VIDEO:** Converting a fraction to a repeating decimal Source: Khan Academy (English - US curriculum)	As a closing individual task (8 minutes), each learner draws the first version of their Personal Number Classification Map in their exercise book. Instructions on the board: 'Draw a large rectangle. Inside it, write the title: WHOLE NUMBERS. Divide the rectangle into sections and place each	Before students begin, display a partially completed EXEMPLAR map on the board — show only the rectangle and the titles, with 2–3 numbers placed, and leave obvious gaps. Say: 'My map is incomplete — yours will be too. That is intentional. A model that is finished on Day 1 means we have	CUMULATIVE MODEL REVISION (Progressive Schematisation): The Personal Number Classification Map is the student's evolving cognitive model of the number system. Starting it in Lesson 1 — deliberately incomplete — establishes a growth mindset around mathematical	OBSERVABLE INDICATOR: Can the learner produce a map that correctly places at least 8 of the 12 numbers into appropriate odd/even and prime/composite categories, with at least one number marked with a question mark or note showing genuine uncertainty? Collect maps at the end of class

🔍 Search ARES for similar videos 📖 **HTML:** Real Number Line Graphs (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	number from 1 to 12 in the correct section(s). Use today's categories: ODD, EVEN, PRIME, COMPOSITE. Some numbers will appear in MORE than one section — that is fine. At the bottom of your map, write one UNRESOLVED QUESTION you still have about how these numbers connect to sharing pizza.' Students are explicitly told: 'This map is NOT finished. You will revise and expand it every lesson for 8 lessons. By Lesson 8, it will also include rational numbers, irrational numbers, and reciprocals.'	nothing left to learn.' Circulate during the 8 minutes. When you see a student place '1' in the PRIME section, quietly say: 'Interesting — write a question mark next to 1 and keep it. We will resolve that in Lesson 2.' When you see a student ask 'Can a number be both odd AND prime?', say: 'Yes — and write that question on your DQB note next lesson. It is a great one.' In the last 2 minutes, ask the class: 'Raise your hand if your map has at least one number you are unsure about.' Normalise uncertainty: 'Good. That means your brain is working. Scientists in Kenya and around the world carry maps with question marks every day.'	understanding. Each subsequent lesson will add a new layer (rational/irrational in Lessons 2–3, reciprocals in Lessons 4–6), making conceptual growth visible and concrete. The physical act of drawing and revising the map is a sensemaking strategy aligned with the NGSS Practice of Developing and Using Models.	(or have students photograph them). Before Lesson 2, identify the 3 most common classification errors and address them in the opening 5 minutes of Lesson 2.

D. TEACHER REFLECTION

After delivering this lesson, reflect on the following questions before planning Lesson 2:

1. PHENOMENON LAUNCH: Did students genuinely find the pizza scenario puzzling, or did it feel artificial? If it felt forced, consider bringing in an actual round object (chapati, a paper plate) to make the 'dividing into thirds' physical and tangible.

2. DQB QUALITY: Read all WONDER notes tonight. Are students asking questions at the right level — about decimal behaviour and number types — or are questions too surface-level (about pizza toppings)? If too surface-level, open Lesson 2 by modelling a deeper question yourself.

3. CLASSIFICATION ERRORS: How many groups placed '1' as a prime number? This is the most common misconception in this lesson. Plan a 5-minute targeted opener for Lesson 2 that directly addresses the special status of 1 without embarrassing the groups who got it wrong.

4. EQUITY CHECK: Were the same 4–5 students dominating all cold-call responses? Review your cold-call tally and ensure different students are targeted in Lesson 2. Specifically note any female learners or learners from non-dominant groups who were silent today — create deliberate on-ramps for them in Lesson 2's group work.

5. MANIPULATIVE MANAGEMENT: Were fraction circles used productively or did they become a distraction? If distraction was an issue, introduce a clear 'manipulatives down, eyes up' signal for future lessons.

6. MODEL MAPS: Scan the collected maps. Which students show strong classification ability and are ready for extension? Which students need additional scaffolding before Lesson 2 introduces rational/irrational distinctions? Plan at least one differentiated task for Lesson 2 based on this data.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Students observed two pizza diagrams — one divided into 4 equal slices (each = 1/4 = 0.25, a terminating decimal) and one divided into 3 equal slices (each = 1/3 = 0.3333…, a never-ending repeating decimal). They also encountered $\sqrt{2} \div 2 \approx 0.7071\ldots$ as a decimal that never ends and never repeats. Using fraction-circle manipulatives, they recorded decimal equivalents for slices of 2, 3, 4, 5, 6, 8, and 10, noting whether each decimal terminates, repeats, or goes on forever.

What did I learn?	Students learned that whole numbers from 1 to 12 can be classified as odd or even (based on divisibility by 2) and as prime or composite (based on the number of factors). They discovered that prime numbers cannot be evenly re-divided into smaller whole-number groups, which connects to why some pizza shares feel 'messy'. They began their Personal Number Classification Map and opened the Driving Question Board with self-generated questions about why some decimals end and others do not.
How does this explain the phenomenon?	The TYPE of whole number used to count pizza slices determines whether each person's share can be written as a 'clean' terminating decimal or produces a decimal that goes on forever. Numbers like 2, 4, 5, 8, and 10 produce terminating decimals; 3, 6, and 7 produce repeating or non-terminating decimals. This is connected to whether the denominator has factors of only 2 and 5 — but students have not yet been told this rule formally. The puzzle of WHY this happens, and what it means for 'irrational' numbers like $\sqrt{2}$, is left open as the driving question for the rest of the unit.

LESSON 2 (80 minutes): Sorting the Number World: Odd, Even, Prime, and Composite — and the Pizza That Started It All	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To classify whole numbers as odd, even, prime, and composite, and to begin connecting these categories to the anchoring phenomenon of why some pizza slices produce neat fractions and others do not.
Knowledge	Learners will know: (a) the definitions of odd, even, prime, and composite numbers; (b) how to distinguish between these four categories using examples drawn from whole numbers; (c) that the type of number used to describe a quantity affects how it behaves in sharing and division contexts.
Skills	Learners will be able to: (a) classify whole numbers as odd, even, prime, and composite in different situations; (b) justify their classification using mathematical reasoning; (c) connect number classification to real-world sharing problems such as the pizza phenomenon.
Attitudes	Learners will appreciate the value of precise classification in mathematics and will collaborate respectfully in peer discussions, reflecting the KICD value of Unity.
Key Inquiry Question	How do we use real numbers in day-to-day activities?
Purpose in Storyline	This is Lesson 2 of 7 in the evidence-gathering sequence. In Lesson 1, students were introduced to the anchoring phenomenon — the Unequal Pizza Slices Problem — and made initial predictions on the Driving Question Board (DQB). In this lesson, students examine the pizza diagram and fraction circles hands-on, then sort whole numbers into odd, even, prime, and composite categories. This categorical thinking is the foundational scaffolding that will later support the classification of real numbers as rational or irrational. The lesson ends with students updating the DQB with new questions sparked by their sorting activity.
Safety Notes	No physical hazards anticipated. Ensure fraction circle manipulatives are stored safely after use. If scissors are used for any cut-out activities, supervise carefully and collect all materials after the lesson.

B. LESSON OVERVIEW
Students begin by revisiting the pizza phenomenon and examining physical fraction circle manipulatives. They make and record predictions about why some pizza cuts produce neat numbers. The teacher then guides a structured peer-discussion activity in which students sort a curated set of whole numbers (2, 3, 4, 9, 15, 17, 36, 41, 49, 64) into odd, even, prime, and composite categories, using a Kenyan market context (counting mandazi on a tray) as an anchor. Students justify their sorting decisions aloud, and the class builds a shared classification wall chart. In the Explain phase, the teacher formalises definitions and links back to the pizza diagram: why does dividing a pizza into 2, 4, or 8 slices feel 'clean', while dividing into 3, 7, or 17 slices feels harder to describe? Students update the Driving Question Board with newly generated questions, then revise their initial class model of the number world.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** Fractions on a	Students sit in groups of four. Each group has a set of fraction circle manipulatives and the pizza	Distribute fraction circles and pizza diagrams before students arrive so materials are ready. Say: 'Before	Think-Pair-Share with DQB Sticky Notes: This strategy activates prior knowledge and externalises	Observable indicator: Every student has written at least one prediction on a sticky note and

number line Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Real Number Line Graphs (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	phenomenon diagram (two pizza images: one cut into 4 equal slices labelled 1/4 each, one cut into 3 slices with an arc length described as approximately 0.333… or √2 ÷ 2 ≈ 0.7071…). Students silently examine the diagrams for 2 minutes, then respond in writing to the prompt: 'Look at the two pizza diagrams. Why do you think one slice can be written as a neat fraction and the other cannot? What is special about the NUMBER of slices?' Students share predictions with a partner (Think-Pair-Share), then two or three pairs share with the class. Students record their predictions on a sticky note and place it on the Driving Question Board (DQB) under the column 'What we think we know'.	we learn anything new today, I want you to think like a mathematician. Look at these two pizzas. One is divided into 4 slices, one into 3 slices. Write down — by yourself, no talking — why you think one produces a neat number and the other does not. You have 2 minutes.' [Allow 2 minutes of silent writing.] Then say: 'Now turn to your partner. Share your prediction. You have 90 seconds.' [Allow 90 seconds.] Cold-call 3 pairs using a name-stick jar — select one strong prediction, one partial prediction, and one misconception. For each, ask: 'Can someone build on that idea?' [WAIT TIME: 12 seconds after each cold call before prompting.] Do NOT correct misconceptions yet — affirm all contributions with 'That's an interesting hypothesis — let's keep it on the board and test it today.' Direct students to write predictions on sticky notes and post them on the DQB.	student thinking before instruction begins. By posting predictions publicly on the DQB, students are held accountable to their initial ideas and will visibly revise them as evidence accumulates across the 7-lesson sequence. This mirrors how scientists record hypotheses before experimentation.	posted it on the DQB. The teacher scans sticky notes during pair-share and mentally notes 2–3 key misconceptions to address later (e.g., 'irrational numbers are wrong answers', 'all decimals are irrational'). No grading — purely diagnostic.
Observe Phase 📹 **VIDEO:** Fractions on a number line Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Real Number Line Graphs (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Students receive a sorting card set: 10 number cards printed with the numbers 2, 3, 4, 9, 15, 17, 36, 41, 49, 64. They also receive a sorting mat with four labelled columns: ODD | EVEN | PRIME | COMPOSITE. The teacher frames the activity with a Kenyan market context: 'Imagine you are at Gikomba Market in Nairobi, and you are arranging mandazi on trays. You have these quantities of mandazi. Which quantities can you arrange in exactly 2 equal rows with none left over? Which can only be arranged in one row?' Groups physically place each number card in what they believe is the correct column, discussing in Swahili or English. After 8 minutes	Circulate during the 8-minute group sort. Do NOT give answers. Instead, ask probing questions: 'How many factors does 9 have? Can you list them?' or 'You said 15 is prime — can you show me that 15 has no factors other than 1 and 15?' [WAIT TIME: 10 seconds after each probe before offering a hint.] During Gallery Walk, narrate: 'I am seeing some interesting disagreements about 49 and 1. That's exactly the kind of thinking we need.' After groups return, cold-call 4 groups — one per category — to justify one of their placements: 'Group 3, tell me why you placed 36 in the COMPOSITE column. Give me at least two factors.' [WAIT TIME: 12 seconds.]	Card Sort with Gallery Walk: Physical manipulation of number cards requires students to operationalise definitions, not just recall them. The Gallery Walk introduces productive disagreement — students must compare their reasoning with peers', which surfaces misconceptions in a low-stakes, collaborative way. This mirrors the NGSS practice of Obtaining, Evaluating, and Communicating Information.	Observable indicator: Teacher checks each group's sorting worksheet. Success criterion: at least 8 of 10 numbers correctly classified. Specific watch-points: (a) Is 9 classified as composite (correct) or prime (misconception)? (b) Is 1 classified as prime (misconception) or neither (correct)? (c) Is 2 identified as the only even prime? Record group-level errors on a clipboard for targeted follow-up in the Explain phase.

	of group sorting, groups do a 'Gallery Walk' — they circulate to see how other groups sorted and use a sticky dot to mark any card placement they disagree with. Students return to their seats and reconcile any disagreements within their group. Each group records final answers on a sorting worksheet.	If a group places 1 in the PRIME column, do not correct immediately — say: 'Interesting — keep that thought. We'll come back to the number 1 in our Explain phase.'		
Explain Phase 📹 **VIDEO:** [Fractions on a number line](#) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Real Number Line Graphs (Flexbook)](#) Source: CK-12 🔍 Search ARES for similar readings	The teacher leads a whole-class debrief using the shared sorting wall chart (a large poster version of the sorting mat). Definitions are formalised: Odd numbers (not divisible by 2), Even numbers (divisible by 2), Prime numbers (exactly two distinct factors: 1 and itself), Composite numbers (more than two factors). The teacher addresses the 'number 1' edge case explicitly. Students then apply their understanding to the pizza phenomenon: 'If a pizza is cut into a PRIME number of slices — like 3, 5, or 7 — can you always share it equally between 2 people without cutting further? What about a COMPOSITE number of slices, like 4, 6, or 8?' Students discuss in pairs for 3 minutes, then the class debriefs. Students independently complete a short written task: classify 6 new numbers and write one sentence connecting their classification to the pizza phenomenon.	Stand at the sorting wall chart. Say: 'Let's build our class definition together. I'm going to point to each column. Tell me: what do all the numbers in this column have in common?' [WAIT TIME: 12 seconds per column before calling on a student.] Cold-call 4 students — one per column — to propose a definition in their own words. Write student-language definitions on the board, then refine to formal KICD definitions. For number 1, say: 'Some of you placed 1 in the PRIME column. Here's the debate mathematicians had for centuries — and here's the answer we use today: 1 is NEITHER prime NOR composite. It has only one factor — itself. Write that in your notes.' Transition to pizza connection: 'Now look at the pizza diagram again. The pizza with 4 slices — is 4 prime or composite?' [Cold-call 1 student.] 'The pizza with 3 slices — prime or composite?' [Cold-call 1 student.] 'Does being prime or composite change how easily we can share? Discuss with your partner for 3 minutes.' [Circulate, listen, collect 2 interesting responses to share.]	Collaborative Definition Building + Phenomenon Reconnection: By having students propose definitions in their own words before the teacher formalises them, the explain phase becomes generative rather than passive. Reconnecting each definition to the pizza phenomenon keeps the sensemaking anchored to a concrete, memorable context — students are not learning isolated vocabulary but building a tool for understanding why the pizza problem works the way it does. This reflects the NGSS practice of Constructing Explanations.	Observable indicator: Students complete the 6-number classification task independently. Success criterion: at least 5 of 6 correctly classified with a valid sentence connecting classification to pizza sharing. The teacher collects these slips at the end of the phase to identify students who need re-teaching before Lesson 3. Additionally, listen for students who spontaneously use the word 'factors' in their pizza explanations — this indicates readiness for the reciprocal concept in Lesson 3.
Driving Question Board (DQB) Creation	Students return to the DQB. The teacher reveals the three DQB columns: 'What we PREDICTED' | 'What we have LEARNED' | 'What	Say: 'We've now sorted numbers into four types. But look back at our original pizza question: WHY can some slices be written as neat	DQB Wonder-Question Protocol: Publicly tracking 'what we still wonder' prevents the false closure that often follows a successful	Observable indicator: Every student posts at least one wonder-question on the DQB. The teacher photographs the DQB at the end of

📹 **VIDEO:** Fractions on a number line Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Real Number Line Graphs (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	we still WONDER'. Students individually write one new question on a sticky note — something they still wonder about after today's lesson — and post it in the 'What we still WONDER' column. The class reads 4–5 wonder-questions aloud. Students also move any sticky note from 'What we PREDICTED' to 'What we have LEARNED' if today's lesson confirmed or corrected it. The teacher highlights 2 wonder-questions that will be answered in coming lessons (e.g., 'Are fractions always rational?' or 'What happens when the pizza slices need irrational numbers?') and labels them with a star: 'We'll come back to these — keep them in mind.'	fractions and others cannot? Have we fully answered that yet?' [Pause — allow students to shake their heads or respond.] 'Good — we haven't. That's the beauty of inquiry. Write down one question you still have. It can start with: Why…? What if…? Can…? How does…? You have 2 minutes — silent writing.' [Allow 2 minutes.] Cold-call 5 students to read their wonder-questions aloud. For each, say: 'Great question — let's post it.' Physically circle 2 questions that connect to upcoming lessons and say: 'These two questions are going to drive our next three lessons. Photograph the DQB with your phone or copy it into your notebook.'	lesson. It models the epistemic practice of science and mathematics — new knowledge generates new questions. Separating 'predicted' from 'learned' also reinforces metacognitive awareness: students see their own thinking change in real time across the unit.	each lesson to track the evolution of class thinking across the 7-lesson sequence. Quality indicator: at least 30% of wonder-questions move beyond classification (e.g., reference fractions, decimals, or sharing — signalling readiness for rational/irrational concepts in Lessons 3–4).
Model Building Phase 📹 **VIDEO:** Fractions on a number line Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Real Number Line Graphs (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each group of four receives a blank 'Number World Map' template — a large circle divided into regions, representing the number system. Today, students label and populate the outer regions with the four categories they have learned: ODD, EVEN, PRIME, COMPOSITE. They place representative numbers from today's card sort into each region and draw arrows to show relationships (e.g., '2 is BOTH even AND prime — it sits on the boundary'). Each group also annotates their map with one unresolved question from the DQB. Students present their maps to an adjacent group in a 2-minute pair-group share. The teacher explains that this map will be revised and expanded in every subsequent lesson — by Lesson 7, it will include rational and irrational numbers, and reciprocals.	Distribute the Number World Map template. Say: 'This map is going to be your growing model of the number world. Think of it like a map of Kenya — right now we've only explored one county. By the end of this unit, we'll have mapped the whole country.' [Brief pause for the metaphor to land.] 'Place the numbers from today's card sort into the correct regions. If a number belongs to more than one region — like 2 — show that with an overlap or an arrow.' [Allow 8 minutes of group work — circulate and prompt with: 'Can a number be both prime and even at the same time? Show me where 2 goes.' WAIT TIME: 10 seconds.] After pair-group share, cold-call 2 groups to describe one interesting placement decision. Close with: 'Leave space in the centre of your map. In the next lesson, we'll start filling in something even bigger — ALL real numbers.'	Cumulative Model Revision (Number World Map): A living, revisable model gives students a spatial, visual representation of the number system that grows with their understanding. Annotating the model with unresolved questions embeds the DQB directly into students' artefacts, reinforcing that models are always incomplete and subject to revision — a core NGSS Science and Engineering Practice (Developing and Using Models). The Kenya map metaphor grounds the abstract idea of a 'number system' in a familiar geographic frame.	Observable indicator: Each group's Number World Map correctly places at least 8 of the 10 sorted numbers from the card sort, with at least one annotated relationship (e.g., an arrow or overlap showing 2 as even and prime). The teacher collects or photographs maps. Watch for groups who draw rigid, non-overlapping regions — this signals a misconception about mutual exclusivity of categories (prime vs. odd) that must be addressed at the start of Lesson 3.

D. TEACHER REFLECTION
After the lesson, reflect on the following questions: (1) Did all groups correctly distinguish prime from composite, particularly for numbers like 9 (= 3 × 3) and 49 (= 7 × 7)? If not, plan a 5-minute warm-up for Lesson 3 using factor trees. (2) Were the wonder-questions on the DQB mostly about classification (surface level) or did they begin to probe WHY some numbers produce neat fractions (deeper level)? If mostly surface, consider adding a short provocative demonstration at the start of Lesson 3 — e.g., dividing 1 by 3 on a calculator and showing the repeating decimal. (3) Did any students conflate 'odd' with 'prime'? This is a common misconception (e.g., 9 is odd but not prime). Note these students for targeted support. (4) Was the Kenyan market mandazi context (Gikomba Market) effective in grounding the abstract sorting task? If students were disengaged during the card sort, consider using a more familiar local context in Lesson 3 — e.g., dividing githeri portions at a school canteen. (5) Were all groups able to complete the Number World Map in 8 minutes? If not, consider reducing the number of card sort items from 10 to 6 in Lesson 3's warm-up revision.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Students examined two pizza diagrams (one with 4 equal rational slices, one with 3 slices described by a non-terminating decimal) and physically sorted 10 number cards (2, 3, 4, 9, 15, 17, 36, 41, 49, 64) into odd, even, prime, and composite categories using a sorting mat. They also completed a Gallery Walk to compare group sorting decisions.
What did I learn?	Whole numbers can be classified as odd (not divisible by 2), even (divisible by 2), prime (exactly two factors: 1 and itself), or composite (more than two factors). The number 1 is neither prime nor composite. The number 2 is uniquely both even and prime. The type of number used to describe pizza slices affects how easily equal sharing can be expressed — prime-slice pizzas are harder to share equally between 2 people without further cutting.
How does this explain the phenomenon?	Students connected number classification to the anchoring pizza phenomenon: when a pizza is cut into a prime number of slices (e.g., 3 or 7), equal sharing between 2 people always requires further division, while composite-slice pizzas (e.g., 4 or 8) allow clean equal sharing. Students posted wonder-questions on the DQB and began constructing a cumulative Number World Map that will be revised across all 7 lessons to eventually include rational and irrational numbers.

LESSON 3 (80 minutes): Reciprocals of Real Numbers — Division, Tables, and Calculators

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To enable learners to determine and apply reciprocals of real numbers using division, mathematical tables, and calculators, and to connect the concept of reciprocals to the pizza-sharing phenomenon — asking: if we know the whole pizza and the number of slices, can we always find how many slices fit into the whole?
Knowledge	Learners will know that: (c) the reciprocal of a real number n is 1/n, found by division; (d) reciprocals of real numbers can be read from mathematical tables and computed using calculators; (e) reciprocals can be applied in mathematical computations including problems involving sharing and measurement.
Skills	Learners will be able to: work out reciprocals of real numbers by division; read reciprocal values from mathematical tables; use a calculator to determine reciprocals; apply reciprocals to solve real-world mathematical tasks connected to the pizza-sharing phenomenon.
Attitudes	Learners will: (f) promote the use of real numbers in day-to-day activities; (Value — Responsibility) take care of calculators and mathematical tables when working out reciprocals; (Value — Unity) collaborate in groups to categorise and compute reciprocals.
Key Inquiry Question	How do we use real numbers in day-to-day activities?
Purpose in Storyline	In Lesson 1, learners classified whole numbers (odd, even, prime, composite). In Lesson 2, learners placed rational and irrational numbers on the number line and distinguished their decimal expansions. Now in Lesson 3, learners confront the reciprocal — the mathematical operation that asks 'how many of this piece fit into the whole?' They apply this to both rational numbers (e.g. 1/4 of a pizza → reciprocal is 4 slices) and irrational numbers (e.g. a slice of size √2/2 → reciprocal is √2 ≈ 1.414…), discovering that reciprocals exist for all non-zero real numbers but behave differently depending on number type. This advances the driving question by showing that knowing a number's type predicts whether its reciprocal is neat or messy.
Safety Notes	No physical hazards. Remind learners to handle calculators and printed mathematical tables carefully — return them undamaged. Ensure shared calculators are passed gently between group members.

B. LESSON OVERVIEW

Learners return to the Unequal Pizza Slices anchoring phenomenon and ask a new question: if each person's slice is a known fraction of the pizza, how many such slices make up the whole? This is the reciprocal. Through a three-round evidence-gathering sequence — (1) computing reciprocals by hand division, (2) reading them from mathematical tables, and (3) verifying with calculators — learners build fluency with all three methods. They apply reciprocals to both rational slice sizes (1/4, 2/3, 3/8) and irrational slice sizes (√2, π/4), noticing that the reciprocal always exists for non-zero real numbers but that irrational reciprocals remain irrational. The lesson uses a Predict–Observe–Explain structure, with pair work, guided group discussion, and a gallery walk of worked examples. Sensemaking is anchored by a 'Reciprocal Number Line Poster' that learners build incrementally and a comparison table of rational vs. irrational reciprocals.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase	Learners are shown two pizza	Display the two pizza diagrams	Prediction Anchoring — By	Observable indicator: Learners

📹 **VIDEO:** [Finding the Reciprocal of Fractions] Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** [Identification and Writing of Reciprocal Fractions (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	diagrams on the board (same as Lesson 1 anchor): Pizza A divided into 4 equal rational slices (each = 1/4), and Pizza B where one slice has an arc length corresponding to √2/2 of the radius. The teacher poses the Predict prompt: 'Each slice of Pizza A is 1/4 of the whole. Without calculating — how many of those slices fit into the whole pizza? Write your prediction as a number. Now look at Pizza B — one slice is approximately 0.7071 of the pizza. How many of THOSE slices do you predict fit into the whole? Write your prediction.' Learners write individual predictions in their exercise books (2 minutes), then share with a shoulder partner (1 minute). Selected pairs share predictions aloud. Teacher records a range of student predictions on the board without correcting them, labelling the column 'Our Predictions'.	prominently. Say: 'We are back to our Nairobi pizza problem. Last lesson we placed these numbers on a number line. Today we ask a new question — how many slices fill the WHOLE pizza?' Distribute the Predict Slip (a half-sheet with the two questions). After 2 minutes of silent individual writing, say: 'Turn and tell your partner your prediction for Pizza B. You have 60 seconds — go.' After partner talk, cold-call 4 pairs (choose 2 who look confident, 2 who look uncertain). Ask each: 'What did you predict and WHY?' WAIT TIME: 12 seconds after each response before commenting. Record all predictions verbatim on the board. Do NOT affirm or correct yet. End by saying: 'Hold these predictions. We are about to collect evidence.'	committing to a written prediction before instruction, learners activate prior knowledge (division, fractions) and create a cognitive need to resolve the tension between Pizza A (intuitive answer: 4) and Pizza B (non-intuitive answer: ≈1.414). This sets up productive dissonance that drives engagement during the Observe phase.	write a numerical prediction for both pizza cases within 2 minutes. Listen for whether learners connect 'how many fit' to division or multiplication. Flag any learner who writes '0' or 'infinite' for Pizza B — they may hold a misconception that irrational numbers cannot be divided. Use cold-call responses to gauge whether learners already intuit the reciprocal concept (e.g. 'I said 4 because 4 × 1/4 = 1').
Observe Phase 📹 **VIDEO:** [Finding the Reciprocal of Fractions] Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** [Identification and Writing of Reciprocal Fractions (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Evidence is gathered in three sequential rounds, each lasting approximately 10–12 minutes. ROUND 1 — Division by Hand: Learners work in pairs. Each pair receives a Reciprocal Worksheet listing six real numbers: 1/4, 2/3, 3/8, 5, √4 (=2), and 0.25. They compute the reciprocal of each by division (reciprocal of n = 1 ÷ n), recording both the fraction form and the decimal form. They check: reciprocal × original = 1. ROUND 2 — Mathematical Tables: Learners receive printed reciprocal tables (or textbook pages). They locate the reciprocal of each number from Round 1 in the table, record the table value, and compare it to their hand-computed answer. They note any rounding differences. ROUND 3 — Calculator: Learners use the	Before Round 1, say: 'The reciprocal of a number n is simply 1 divided by n. It answers the question: how many of this piece fit into 1 whole? Let's compute.' Circulate during Round 1. Stop at pairs struggling with fractions — ask: 'What is 1 ÷ (2/3)? Remember, dividing by a fraction means multiplying by its flip.' WAIT TIME: 10 seconds. Before Round 2, model ONE table lookup on the board using the number 4: 'Find 4.000 in the left column — read across to get 0.2500. Does that match your division answer?' Before Round 3, demonstrate the 1/x key on one calculator projected or held up. Say: 'Now enter √2. What do you get? Now press 1/x. Record every digit you see.' Cold-call 3 learners to read their	Three-Method Triangulation — By computing the same reciprocal three ways (division, tables, calculator), learners verify answers through convergence. Discrepancies (e.g. rounding in tables) become data, not errors — learners see why calculators are more precise. This strategy builds procedural fluency alongside conceptual understanding, and keeps all three methods visible for the summative skill outcomes (c), (d), and (e).	Observable indicator: Learners' four-column comparison table is complete with all six numbers from Rounds 1–2 and both irrational numbers from Round 3. Check for three specific errors: (1) writing the reciprocal of 2/3 as 2/3 rather than 3/2 (confusion of reciprocal with the number itself); (2) leaving √2 reciprocal as a fraction 1/√2 without rationalising or converting to decimal; (3) failing to verify reciprocal × original = 1. Circulate and stamp or initial correct tables before Phase 3 begins.

	classroom calculators (or phones where available) to compute reciprocals using the 1/x or x^{-1} key. They compute reciprocals of two additional numbers: √2 and π/4. They record all decimal places shown on the display and note whether the decimal terminates, repeats, or continues without pattern. All results are recorded in a four-column comparison table: Number | Reciprocal by Division | Reciprocal from Tables | Reciprocal from Calculator.	calculator displays aloud for √2 reciprocal. Ask: 'Is this decimal terminating, repeating, or neither? How do you know?' WAIT TIME: 15 seconds. After Round 3, say: 'Compare all three columns of your table. What do you notice about the three methods?' Give 90 seconds of silent comparison time before moving to Explain.		
Explain Phase 📹 **VIDEO:** Finding the Reciprocal of Fractions Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Identification and Writing of Reciprocal Fractions (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Groups of four convene. Each group receives a 'Reciprocal Sorting Card Set' — 12 cards showing numbers (including fractions, integers, surds, and decimals) and their reciprocals, some matched, some not. Task 1 (6 min): Sort cards into correct number–reciprocal pairs. Task 2 (4 min): Classify each reciprocal as rational or irrational. Task 3 (5 min): Answer the written question on the card set — 'A mandazi seller in Kisumu sells each mandazi for KSh 1/3 of her daily target. How many mandazi must she sell to reach her daily target? Write this as a reciprocal calculation.' (Answer: reciprocal of 1/3 = 3 mandazi.) Task 4 (4 min): Groups write a one-sentence rule on a sticky note: 'The reciprocal of a rational number is always _____, and the reciprocal of an irrational number is always _____.' Groups post sticky notes on the class 'Sensemaking Wall'. Teacher facilitates a 5-minute gallery walk where learners read each other's rules and place a tick if they agree or a question mark if they disagree.	Distribute Sorting Card Sets face-down. Say: 'Flip them over — you have 6 minutes to match every number to its reciprocal. Use your table from the Observe phase.' Circulate; do not give answers. When a group finishes early, prompt: 'Now classify each reciprocal — rational or irrational. What pattern do you see?' After Task 2, cold-call one group to share their classification. Ask the class: 'Does every group agree? Thumbs up, sideways, or down.' Address thumbs-sideways before continuing. For the mandazi problem, say: 'Think about a mandazi seller in Kisumu. If each mandazi is 1/3 of her target, how many does she sell to hit her goal? Don't just say 3 — write the reciprocal calculation that proves it.' WAIT TIME: 12 seconds. After the gallery walk, bring the class back and read two contrasting sticky notes aloud. Ask: 'Which rule is more precise — and why?' Establish the class consensus rule verbatim before moving on.	Card Sort with Collaborative Rule Construction — The physical sorting task externalises learners' reasoning and makes misconceptions visible (e.g. pairing √2 with √2 instead of 1/√2). Writing the one-sentence rule requires learners to synthesise the pattern across all examples into transferable language. The gallery walk creates peer accountability — learners must justify their rule to others, deepening retention and surfacing remaining uncertainties.	Observable indicator: (1) All 12 card pairs correctly matched — any mismatch identifies a procedural gap. (2) Correct classification of reciprocals as rational/irrational — listen for the rule that 'the reciprocal of a rational non-zero number is rational; the reciprocal of an irrational number is irrational.' (3) Mandazi problem solved as a reciprocal calculation (not just intuition). (4) Sticky note rule is mathematically accurate and uses the words 'rational' and 'irrational'. Flag groups whose rule omits the condition 'non-zero' — address in Phase 4.

Driving Question Board (DQB) Creation 📹 **VIDEO:** [Finding the Reciprocal of Fractions] Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** [Identification and Writing of Reciprocal Fractions (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	The class Driving Question Board (a large poster or section of the board preserved across all 7 lessons) is displayed. Learners individually write on sticky notes: one GREEN note — 'Something I now understand about the pizza problem that I didn't before', and one YELLOW note — 'A new question this lesson raised for me.' They post notes on the DQB under the columns 'We Now Know' and 'We Still Wonder'. Teacher reads 3–4 notes aloud and connects them explicitly to the driving question: 'Why can some numbers be written as exact fractions and others cannot — and how does knowing a number's type help us solve everyday problems in Kenya?' The teacher then highlights one yellow note that will be answered in Lesson 4 (e.g. 'What happens when you take the reciprocal of 0?' or 'Can you use reciprocals to solve equations?'), creating an anticipation bridge.	Distribute two sticky notes per learner. Say: 'Green note — one thing you NOW understand about our pizza problem that you didn't understand at the start of today's lesson. Yellow note — one NEW question today's lesson raised for you. You have 90 seconds. Go.' After posting, read 2 green notes and 2 yellow notes aloud. For each green note ask the class: 'Does this match what YOU wrote? Thumbs up if yes.' For yellow notes, say: 'This question — [read it] — is a great one. Here is a hint about where we are heading in Lesson 4…' Deliberately leave the hint incomplete to build anticipation. Finally, point to the original prediction column on the board from Phase 1 and ask: 'Look at our Pizza B prediction. Who was right? Who wants to revise?' Cold-call 2 learners to compare prediction to evidence. WAIT TIME: 10 seconds each.	Driving Question Board Anchoring — The DQB makes the growth of understanding visible across lessons. By physically posting 'We Now Know' evidence against the driving question, learners experience the incremental nature of scientific and mathematical inquiry. Yellow notes preserve productive uncertainty, which is crucial for maintaining curiosity across the remaining 4 lessons in the sequence.	Observable indicator: GREEN notes reference the reciprocal concept specifically (not vague statements like 'I learned maths'). YELLOW notes raise genuine mathematical questions (not social comments). Teacher checks that at least one green note per table group explicitly connects reciprocals to the pizza phenomenon. If green notes are too vague, prompt: 'Can you say HOW reciprocals connect to our pizza problem before you post?'
Model Building Phase 📹 **VIDEO:** [Finding the Reciprocal of Fractions] Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** [Identification and Writing of Reciprocal Fractions (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Learners return to their individual 'Pizza Model Booklet' — a folded A4 booklet begun in Lesson 1 that they update each lesson. Today's model-building task has two parts. Part A (5 min): Learners add a 'Reciprocal Layer' to their pizza diagram — for Pizza A (rational slices of 1/4), they annotate: 'Reciprocal = 4; meaning: 4 slices fill the whole; type: rational.' For Pizza B (irrational slice of √2/2), they annotate: 'Reciprocal = √2 ≈ 1.414…; meaning: approximately 1.414 of these slices fill the whole; type: irrational — and the reciprocal is also irrational.' Part B (4 min): Learners add a 'Methods Box' below their diagram listing all three methods for finding a	Distribute model booklets. Say: 'Open to your pizza diagram. You are going to add a new layer today — the Reciprocal Layer. Look at the board — I will show you exactly what to annotate for Pizza A.' Sketch the annotation live on the board. Then say: 'Now YOU annotate Pizza B using what you found in the Observe phase. Use your calculator results.' Circulate for 4 minutes. Stop at 2–3 learners and ask: 'Read me your sentence for the driving question connection.' WAIT TIME: 10 seconds. Prompt if vague: 'Can you use the words rational, irrational, and reciprocal in that sentence?' In the last 90 seconds, say: 'Hold up your Methods Box so	Cumulative Model Revision — The Pizza Model Booklet functions as a growing, learner-generated scientific model. Each lesson adds a new explanatory layer, making conceptual growth tangible. Annotating both pizza cases (rational and irrational) side-by-side forces comparison thinking and prevents learners from treating the concepts in isolation. The driving-question sentence in Part C is a micro-writing task that consolidates language and conceptual integration.	Observable indicator: Part A annotations are mathematically correct — specifically, the reciprocal of √2/2 is correctly identified as 2/√2 = √2 (rationalised) ≈ 1.414, not 0.7071. Part B Methods Box contains all three methods with a worked example for each. Part C sentence uses domain vocabulary (reciprocal, rational, irrational) and references the pizza phenomenon. Collect booklets and use a 3-point rubric: 3 = all three parts complete and accurate; 2 = two parts complete; 1 = one part complete. Return with written feedback before Lesson 4.

	reciprocal (division, tables, calculator) with one worked example each, drawn from today's lesson. Part C (2 min): Learners write one sentence connecting today's model update to the driving question. Model booklets are collected by the teacher for formative review and returned in Lesson 4.	your neighbour can see it. Does their box match yours? Give them one star and one suggestion on a sticky note.' Collect booklets at the door as an exit task.		

D. TEACHER REFLECTION
After the lesson, reflect on the following questions: (1) PROCEDURAL FLUENCY — Were learners able to compute reciprocals using all three methods (division, tables, calculator) with reasonable accuracy? If the majority struggled with mathematical tables, consider dedicating 10 minutes at the start of Lesson 4 to a table-reading drill. (2) CONCEPTUAL DEPTH — Did learners articulate WHY the reciprocal of an irrational number is also irrational? If this reasoning was absent from sticky notes and model booklets, introduce a short proof-by-contradiction scaffold in Lesson 4. (3) PHENOMENON CONNECTION — Did learners use the pizza context naturally in their explanations, or did they treat it as decorative? If the phenomenon felt disconnected, revisit the Pizza B diagram at the start of Lesson 4 and explicitly re-anchor. (4) EQUITY CHECK — Which learners did NOT speak during cold-calls across all three phases? Intentionally target those learners for cold-call inclusion in Lesson 4. (5) MODEL BOOKLET REVIEW — From the collected booklets, identify the most common error (likely: confusing reciprocal of $\sqrt{2}/2$ with 0.7071 rather than $\sqrt{2} \approx 1.414$) and design a targeted correction activity for the opening of Lesson 4.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners computed reciprocals of rational numbers (1/4, 2/3, 3/8, 5, 0.25) and irrational numbers ($\sqrt{2}$, $\pi/4$) using three methods: long division, mathematical reciprocal tables, and calculators. They recorded and compared results across all three methods in a four-column comparison table. They also observed that the calculator display for irrational reciprocals produces non-terminating, non-repeating decimals.
What did I learn?	The reciprocal of any non-zero real number n is 1/n — it answers the question 'how many of this piece fit into 1 whole?' The reciprocal of a rational number is always rational (e.g. reciprocal of 2/3 is 3/2). The reciprocal of an irrational number is always irrational (e.g. reciprocal of $\sqrt{2}$ is $1/\sqrt{2} = \sqrt{2}/2 \approx 0.7071$). All three methods — division, tables, and calculators — yield the same value, with calculators providing the greatest precision. Reciprocals have real-world applications, such as the mandazi seller problem in Kisumu.
How does this explain the phenomenon?	Because we know whether a number is rational or irrational (Lesson 2), we can now predict in advance whether its reciprocal will be neat (a fraction) or messy (an endless non-repeating decimal). This means that for the Nairobi pizza problem: if the slice size is rational, the number of slices that fill the whole is also a rational number — clean and predictable. If the slice size is irrational, the number of slices that fill the whole is also irrational — it exists, but it cannot be expressed as a neat fraction. Knowing a number's type therefore gives us real predictive power about the answers we will get in everyday calculations.

LESSON 4 (80 minutes): Rational or Irrational? Placing Real Numbers on the Number Line	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Students classify real numbers as rational or irrational, place them accurately on a number line, and connect each category back to the pizza-sharing phenomenon — identifying which slice sizes can be written as exact fractions and which cannot.
Knowledge	b) classify real numbers as rational and irrational in different situations.
Skills	Critical Thinking and Problem Solving: the learner uses the reciprocal of real numbers to work out mathematical tasks. Communication and Collaboration: the learner discusses with peers to identify and categorise whole numbers as odd, even, prime, composite, rational, and irrational.
Attitudes	Unity: the learner discusses in a group and categorises real numbers as rational and irrational numbers. Life Skills (self-esteem): the learner participates in the discussion on the concepts of rational and irrational numbers.
Key Inquiry Question	How do we use real numbers in day-to-day activities?
Purpose in Storyline	This is Lesson 4 of 7 in the evidence-gathering sequence. Students have previously classified whole numbers (odd, even, prime, composite) and explored fraction notation. Today they zoom out to the full real number system — placing √2, π, and 0.333… on a number line, attempting to express them as fractions, and discovering that some numbers resist exact fractional representation. This directly advances the driving question by giving students the conceptual vocabulary (rational vs. irrational) to explain WHY two pizza slices that look identical can behave so differently when described with numbers. The DQB gains the first formal entries of 'rational number' and 'irrational number'.
Safety Notes	No hazardous materials are used. Ensure scissors (if used to cut number-line strips) are handled carefully. All printed materials should be kept flat to avoid tripping hazards in group work areas.

B. LESSON OVERVIEW

Learners begin by predicting whether √2, π, and 0.333… can be written as fractions. They then gather evidence through three structured activities: (1) a long-division/decimal-expansion investigation on printed number strips, (2) a group number-line placement exercise using fraction circle manipulatives as benchmarks, and (3) a guided reading excerpt comparing rational and irrational decimal patterns. Through class discussion, they build a formal two-column classification chart, connect each number type to a pizza-slice diagram, and update the Driving Question Board with new vocabulary. The lesson closes with a model-revision task in which students annotate their cumulative pizza diagram to label rational and irrational regions.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** [Prime numbers] Source: Khan	Students receive a 'Pizza Prediction Card' showing four numbers: 1/3, 0.75, √2, and π. They are asked to answer	Distribute Pizza Prediction Cards face-down. Say: 'Before we learn anything new today, I want to see what you already think. Flip your	Predict-Before-Instruction (Elicitation): activates prior knowledge about fractions from Lessons 1–3, surfaces	Observable indicator: Student prediction cards collected at the end of the phase. Teacher scans for: (a) correct identification of 1/3

Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Real Number Line Graphs (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	individually in their exercise books: 'Which of these numbers could describe an exact pizza-slice size that any baker in Nairobi could cut perfectly and label with a simple fraction? Circle your choices and explain your reasoning in one sentence.' Students also sketch a rough number line from 0 to 4 and mark where they think each number sits, without any instruction.	card over — you have 3 minutes to work completely alone. No talking yet.' Set a visible 3-minute timer. Circulate silently; note 3–4 students who have placed √2 near 1.4 and 2–3 who have placed it between 1 and 2 with no further precision. After timer: 'Pens down. I am going to cold-call five people — not for right answers, but for honest predictions.' Cold-call 5 students by name (aim for gender and seating-area spread). After each response, say: 'Interesting — hold that thought.' Do NOT confirm or correct. Close with: 'We have five different ideas on the board. By the end of today you will know exactly which of these are right — and more importantly, WHY.'	misconceptions (e.g. 'π ≈ 22/7 so it IS a fraction'), and gives students a personal stake in resolving the cognitive dissonance during later phases.	and 0.75 as fraction-expressible; (b) uncertainty or incorrect placement of √2 and π. This informs which misconceptions to address explicitly in the Explain phase. Target: ≥ 60% of students should correctly identify at least one rational number before instruction.
Observe Phase 📹 **VIDEO:** Prime numbers Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Real Number Line Graphs (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Activity A — Decimal Expansion Investigation (20 min): In pairs, students perform or verify long division for 1÷3, 3÷4, 1÷7, and use a calculator (or printed decimal table) to find the first 10 decimal places of √2 and π. They record results in a three-column table: Number | Decimal Expansion (first 10 places) | Pattern? (terminating / repeating / neither). Activity B — Number Line Placement (15 min): Groups of four receive a 1-metre paper number-line strip (0 to 4, marked at whole numbers). They are also given their fraction circle manipulatives from previous lessons. Task: physically place sticky-note labels for 1/3, 3/4, √2, π, and 0.333… on the strip, using the fraction circles as benchmarks for 1/3 and 3/4. They photograph their number line with a phone or tablet (if available) or sketch it in their books. Activity C — Guided Reading (5 min): Students read a 150-word excerpt (provided on a	Activity A — Say: 'With your partner, divide 1 by 3 using long division. Write every remainder. Then use your calculator for √2 and π. You have 10 decimal places to fill. Go.' Allow 10 minutes. At minute 8, circulate and ask targeted pairs: 'What do you notice about the remainders when you divide 1 by 3?' [WAIT 12 seconds for response.] 'What about when the calculator shows √2 — do you see any block of digits repeating?' [WAIT 12 seconds.] Activity B — Say: 'Now your group has a number line and your fraction circles. Use the circles as physical rulers — where does 1/3 of the circle's diameter sit on this line?' Prompt groups who are stuck: 'Your fraction circle for one-third — if the whole pizza has diameter that fits from 0 to 1, where does one-third of that land?' [WAIT 10 seconds.] Cold-call 2 groups to explain their placement of √2: 'Tell me: how did you decide	Data-Table Investigation + Physical Manipulation (Concrete-Representational): the decimal-expansion table makes the repeating/non-repeating distinction visible and concrete. The physical number-line strip connects abstract decimals to spatial magnitude — bridging the pizza diagram (a circular model) to a linear model. The guided reading then provides the verbal/conceptual label for what students have just seen empirically.	Observable indicators: (1) Decimal tables checked for correct long-division remainders for 1÷3 (should see remainder cycling: 1, 10, 1, 10…). (2) Number-line strips: √2 must appear between 1 and 2, closer to 1.5; π must appear between 3 and 4, closer to 3. Flag any group placing π between 3.1 and 3.2 (correct) for praise; redirect any group placing π near 4. (3) Underlined sentences in guided reading: accept any sentence that includes the idea of 'non-repeating, non-terminating'. Teacher notes 3 common errors to address in Explain phase.

	printed card) titled 'Why Some Numbers Are Irrational — A Pattern Story' which explains terminating, repeating, and non-repeating non-terminating decimals using the example of dividing a Nairobi pizza into 6 slices (giving 0.1666…) versus the diagonal of a 1×1 ugali square (giving √2).	WHERE to put root-2 on the line?' Activity C — After reading: 'In your book, underline the one sentence that best answers: what makes a decimal irrational?' Allow 2 minutes silent reading + underlining.		
Explain Phase 📹 **VIDEO:** Prime numbers Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Real Number Line Graphs (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Whole-class construction of a 'Real Number Classification Chart' on the board (or large paper). Students contribute evidence from their tables and number lines. The chart has two main columns — RATIONAL and IRRATIONAL — with sub-rows for definition, decimal behaviour, fraction form (yes/no), and pizza-slice example. Students then complete an 'Exit Ticket Explainer': given three new numbers (0.625, √5, 22/7), they must classify each and justify in writing using the vocabulary 'terminating', 'repeating', 'non-repeating', and 'non-terminating'. Finally, students return to their pizza diagrams from Lesson 1 and label each slice region with R (rational) or I (irrational), justifying one labelling choice in a speech bubble.	Facilitate chart-building: 'I am going to draw two columns. Before I write anything, tell me — from your decimal tables — what was TRUE about 1/3 that was NOT true about √2?' [WAIT 15 seconds. Cold-call 3 students.] Record student language first, then refine: 'You said it repeated — mathematicians call that a repeating decimal. What do we call a number whose decimal NEVER repeats and NEVER ends?' Accept answers, then write the formal definition verbatim: 'An irrational number is a real number that cannot be expressed as a ratio p/q where p and q are integers and q ≠ 0.' Say: 'Copy this definition word for word.' Pause 60 seconds for copying. Then address the 22/7 misconception directly: 'Some of you may have heard that π equals 22/7. I want you to use your calculator RIGHT NOW to divide 22 by 7, and then check the first 8 digits of π from your table. Are they identical?' [WAIT 20 seconds.] Cold-call 2 students: 'What did you find?' Establish: 22/7 ≈ 3.142857… but π = 3.14159265… — they are close but NOT equal. Say: '22/7 is rational. π is irrational. They are neighbours on the number line, but they are different types of numbers — like two pizza slices that look	Co-constructed Anchor Chart + Targeted Misconception Busting: building the chart collaboratively from student evidence (not teacher lecture) ensures conceptual ownership. The 22/7 vs. π calculator check is a high-leverage moment that directly resolves the most common Kenyan classroom misconception about π, tying back to the pizza phenomenon. The pizza-diagram labelling task forces students to apply the new vocabulary to the original anchoring context.	Observable indicators: (1) Exit Ticket Explainer — correct classification: 0.625 = rational (terminating), √5 = irrational (non-repeating non-terminating), 22/7 = rational (repeating). Accept written justifications that use at least two vocabulary terms correctly. Target: ≥ 75% of students classify all three correctly. (2) Pizza diagram labelling — all 'halves', 'quarters', 'thirds' regions should be marked R; any arc-length or diagonal-derived slice should be marked I. Teacher collects 8 random exit tickets to inform next lesson grouping.

		the same size but one can be named exactly and one cannot.'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Prime numbers Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Real Number Line Graphs (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each student writes two sticky notes: (a) one NEW THING I NOW KNOW (green note) — must use the words 'rational' or 'irrational' in a complete sentence connected to the pizza phenomenon; (b) one NEW QUESTION I STILL HAVE (yellow note) — a genuine remaining wonder (e.g. 'Can an irrational number ever be a pizza slice in real life?', 'Is there a number between rational and irrational?'). Students post notes on the class DQB under the columns 'Evidence We Have Gathered' and 'Questions Still Chasing Us'. The class reviews the board together for 5 minutes; the teacher reads 3 green and 3 yellow notes aloud and asks: 'Do we now have enough evidence to fully answer the driving question?' Students respond with thumbs up / flat / down as a quick temperature check.	Say: 'Look at our driving question on the board: WHY can some numbers be written as exact fractions and others cannot — and how does knowing a number's type help us solve everyday problems in Kenya? You now have new evidence. Write your green note first — it must connect to the pizza. Then write your yellow note — what are you still wondering?' Allow 4 minutes for writing. Circulate; if a student's green note says only 'rational means fraction', prompt: 'How does that connect to the pizza slices we started with? Add one more sentence.' After posting: Read 3 green notes aloud with genuine enthusiasm: 'Listen to what [Student A] wrote — they connected rational numbers directly to the 1/4 pizza slice. That is exactly the kind of thinking mathematicians use.' Read 3 yellow notes and say: 'These questions are going to drive our next three lessons. Hold onto them.'	Driving Question Board — Iterative Public Record: the DQB makes student thinking visible and cumulative across all 7 lessons. Distinguishing 'evidence gathered' from 'questions still chasing' reinforces scientific/mathematical reasoning habits. The thumbs-up temperature check gives the teacher rapid whole-class formative data on readiness to advance.	Observable indicators: Green notes must contain (a) the word 'rational' or 'irrational', and (b) a reference to pizza, fractions, or slice size. Yellow notes should be genuine mathematical questions, not social comments. Teacher photographs the DQB at the end of each lesson for tracking conceptual progression across the 7-lesson sequence. Flag any yellow notes that reveal a surviving misconception (e.g. 'Is 22/7 the same as π?') for explicit re-teaching in Lesson 5.
Model Building Phase 📹 **VIDEO:** Prime numbers Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Real Number Line Graphs (Flexbook) Source: CK-12 🔍 Search ARES	Students retrieve their 'Cumulative Pizza Model' — a running annotated diagram started in Lesson 1 and updated each lesson. Today's revision task: (1) Add a COLOUR CODE — shade all rational-slice regions in blue and mark any irrational-measure element (e.g. arc length √2 ÷ 2) in red. (2) Add a NUMBER LINE RIBBON — a small strip glued or drawn below the pizza showing where each slice-size sits on the number line from 0 to π+1, with R/I labels. (3) Write one 'Model Claim' sentence at the bottom: 'Based on	Distribute cumulative models (or direct students to their exercise-book pages). Say: 'Your model has grown each lesson. Today you are adding two things — a colour code and a number line ribbon. You have 8 minutes.' Set timer. At minute 5, say: 'You should be starting your Model Claim sentence now.' After sharing pairs: cold-call 4 pairs to read their claim aloud. Record the 4 versions on the board. Ask the class: 'Which version is most precise? Which one would survive if I gave you a number you had never seen	Cumulative Model Revision (Progressive Formalization): updating the same physical model each lesson builds a visible record of conceptual growth and prevents knowledge from remaining isolated to individual lessons. The colour-coding task forces students to apply classification decisions spatially, while the number-line ribbon connects circular (pizza) and linear representations. The Model Claim sentence is a progressive-formalization move — students generate their own definition from evidence before the	Observable indicators: (1) Colour coding — rational regions (halves, quarters, thirds, eighths) correctly shaded blue; irrational elements (arc length, √2-derived measures) correctly marked red. Redirect any student who shades π blue (common error — π is irrational even though 22/7 is close). (2) Number line ribbon — √2 must appear between 1 and 2; π between 3 and 4. (3) Model Claim — must include 'fraction' or 'p/q' and 'integers'. Teacher collects 6 models at random to give written feedback returned in Lesson 5.

for similar readings	today's evidence, I claim that a pizza slice size is rational if and only if ________.' Students share their model claim with a shoulder partner and negotiate the best wording, then one representative from each pair shares with the class.	before — could you use that claim to classify it?' [WAIT 15 seconds.] Guide class to converge on a claim that includes: 'can be expressed as p/q where p and q are integers and q ≠ 0.' Close by saying: 'In Lesson 5, we are going to use this classification to find reciprocals — and you will discover that rational and irrational numbers behave very differently there too. Bring your models back.'	teacher formalizes it.	

D. TEACHER REFLECTION

After the lesson, reflect on the following questions in your professional journal: (1) Which students successfully distinguished π from 22/7 during the calculator check — and which still appeared uncertain? Plan a 5-minute re-engagement for those students at the start of Lesson 5. (2) Did the physical number-line strip activity reveal any spatial misconceptions (e.g. students placing √2 closer to 2 than to 1.4)? If so, consider opening Lesson 5 with a brief number-sense warm-up using decimal benchmarks. (3) Review the DQB yellow notes photographed at the end of this lesson. Tally the top 3 recurring student questions and sequence Lessons 5–6 to address the most common one first. (4) Were all student groups able to use the fraction circle manipulatives as benchmarks, or did some groups abandon them and rely purely on calculators? Groups that abandoned the concrete tool may need additional bridging between circular and linear representations before the reciprocal concept (Lessons 5–7) is introduced. (5) Consider: did the Uganda context feel authentic to your students, or did you find a more locally resonant pizza-sharing analogy emerging in student talk? Capture any student-generated analogies (ugali cutting, mandazi sharing) and weave them into the Lesson 5 phenomenon re-engagement.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Students observed that the decimal expansions of 1÷3 and 3÷4 either terminate or repeat in a predictable pattern, while √2 and π, when displayed on a calculator to 10 decimal places, show no repeating block and no termination. Students also physically placed these numbers on a 1-metre number-line strip, observing that √2 sits between 1 and 2 (closer to 1.4) and π sits between 3 and 4 (closer to 3.14), visually confirming their magnitude even when their exact fractional form is unknowable.
What did I learn?	Students learned that real numbers are classified as rational (expressible as p/q, integers, q ≠ 0; decimal terminates or repeats) or irrational (cannot be expressed as p/q; decimal never terminates and never repeats). Key examples: 1/3 = 0.333… (rational, repeating), 0.75 (rational, terminating), √2 ≈ 1.41421356… (irrational), π ≈ 3.14159265… (irrational). Students also learned that 22/7, while a common approximation for π, is itself a rational number and is NOT equal to π.
How does this explain the phenomenon?	Students explained the pizza-sharing phenomenon using their new classification: pizza slices described by fractions like 1/4 or 1/8 are rational — a Nairobi baker can label them exactly. Slices whose size is linked to a measurement involving √2 (e.g. the diagonal of a square with side equal to the radius) or to π (arc lengths) are irrational — they exist physically on the pizza but cannot be written as a perfect fraction. The driving question advances: knowing a number's type tells us whether its pizza-slice description is exact or only an approximation.

LESSON 5 (80 minutes): Flipping Numbers: Introducing Reciprocals by Division	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To introduce and apply the concept of reciprocals of real numbers through division, using the anchoring pizza-sharing phenomenon as a concrete context.
Knowledge	The learner determines the reciprocal of real numbers by division (KICD SLO c).
Skills	Critical Thinking and Problem Solving: the learner uses the reciprocal of real numbers to work out mathematical tasks. Communication and Collaboration: the learner discusses with peers to identify and compute reciprocals.
Attitudes	Responsibility: the learner takes care of calculators and mathematical tables when working out the reciprocal of real numbers. Unity: the learner works collaboratively in groups to explore reciprocals.
Key Inquiry Question	How do we use real numbers in day-to-day activities?
Purpose in Storyline	In Lessons 1–4, students classified numbers (odd, even, prime, composite, rational, irrational) and observed that some pizza-slice measurements produce neat fractions while others produce non-terminating, non-repeating decimals. Now, in Lesson 5, students confront a new puzzle from the same phenomenon: if we know the size of one pizza slice, how many of those slices fit into the whole pizza? This is the concept of the reciprocal — 'flipping' the number — and students discover it works for both rational and irrational numbers, deepening their understanding of real numbers and their day-to-day utility.
Safety Notes	No physical hazards anticipated. Ensure students handle fraction circle manipulatives and printed mathematical tables carefully. If calculators are shared, students should handle them with care in line with the KICD value of Responsibility.

B. LESSON OVERVIEW
Students revisit the Unequal Pizza Slices anchoring phenomenon and pose the question: 'If each slice is a known fraction of the pizza, how many slices make a whole?' Through hands-on fraction circle manipulation, guided division tasks, and structured group discussion, learners construct the concept of a reciprocal as 'the number you multiply by to get 1 (the whole pizza).' They compute reciprocals of rational numbers by division, then extend this to simple irrational examples (e.g., √2), connecting findings back to the driving question about number types. The lesson advances the evidence-gathering sequence by giving students their first computational tool — the reciprocal — for working with real numbers in everyday contexts such as sharing food, adjusting recipes, and measuring fabric in Kenyan markets.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** [Finding the Reciprocal of Fractions] Source: CK-12	Students are shown a diagram of a pizza divided into 3 equal slices. Each slice is labelled 1/3. The teacher poses: 'If each slice is 1/3 of the pizza, how many slices would fill the whole pizza exactly?	Display the two pizza diagrams used since Lesson 1 (rational-slice version and irrational-slice version) on the board or ARES screen. Say verbatim: 'Look at Slice A — it is exactly 1/3 of the pizza. Without	Prediction Journaling with Public Record: By writing predictions before instruction and displaying them publicly, students activate prior knowledge about fractions and division, surface	Observable indicator: Every student has a written prediction in their exercise book before discussion begins. Teacher scans books during shoulder-partner talk. Listen for the misconception that

🔍 Search ARES for similar videos 📖 **HTML:** Identification and Writing of Reciprocal Fractions (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Write your prediction and your reasoning in your exercise book before we test it.' Students write individual predictions (1–2 minutes), then share with a shoulder partner. The teacher then escalates: 'Now imagine a slice whose size is √2 ÷ 4 of the pizza. Predict: do you think we can still find how many of these slices fill the whole? Why or why not?' Students record a second prediction.	calculating yet, predict how many of Slice A fit into one whole pizza. Write it down. I will not tell you if you are right yet.' Allow 2 minutes of silent individual thinking. After writing, say: 'Turn to your shoulder partner. You have 60 seconds — share your prediction and one reason.' Cold-call 3 pairs (choose one who predicted correctly, one who predicted incorrectly, one who was unsure) using name cards. Record all three predictions on the board under the heading 'Our Predictions — We will check these.' Do NOT confirm or deny. Then pose the irrational extension question. Allow 1 minute of silent writing. Cold-call 2 students. Record responses. Say: 'Hold these predictions — our investigation today will tell us who is right.'	misconceptions (e.g., 'you can't divide by an irrational number'), and create a personal stake in the outcome. This strategy builds metacognitive awareness — students will return to their predictions at the end of the lesson to self-assess.	'flipping' only works for whole numbers or simple fractions. Note which students cannot articulate a reason — these students need targeted support in Phase 2. Cold-call responses reveal the range of prior understanding.
Observe Phase 📹 **VIDEO:** Finding the Reciprocal of Fractions Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Identification and Writing of Reciprocal Fractions (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Students work in groups of 4 with fraction circle manipulatives. Each group receives a printed Activity Card with three tasks: (1) Use the fraction circles to physically show how many 1/3 pieces fill the whole circle. Record: 3 × (1/3) = 1. (2) Now use division: the whole pizza is 1. Each slice is 1/3. Write the division: 1 ÷ (1/3) = ? Compute this by hand using the rule 'divide by a fraction = multiply by its flip.' Record your answer. (3) Repeat for slice sizes: 1/4, 2/5, and 3/7. Complete the table: Slice Size | 1 ÷ Slice Size | Answer | What do you notice? After the table, students read a short printed scenario: 'Amina runs a mandazi stall in Gikomba Market, Nairobi. Her recipe uses 2/3 kg of flour per batch. She has 1 kg of flour. How many full batches can she make?' Students solve using reciprocal	Distribute fraction circle sets and Activity Cards. Say: 'In your groups, start with Task 1 — use the physical pieces. You have 3 minutes. I will walk around and I want to see the circles on your desks, not packed away.' Circulate for 3 minutes. After Task 1, say: 'Now Task 2. Use the division rule we discussed in earlier lessons about rational numbers — if you are unsure, look at the Rational Numbers anchor chart on the wall.' Allow 5 minutes for Tasks 2 and 3 (the table). After groups complete the table, ask: 'What pattern do you see in your table? Discuss in your group for 90 seconds.' Cold-call 2 groups to share patterns. Anticipated response: 'The answer is always the flipped version of the fraction.' Respond: 'Interesting — so what is the relationship between the slice size and the	Concrete-Representational-Abstract (CRA) Progression: Students begin with physical fraction circles (Concrete), then record the relationship in a table (Representational), then abstract the rule as 'dividing by a number gives its reciprocal' (Abstract). This progression is grounded in the pizza phenomenon — the fraction circles ARE the pizza slices — so abstraction emerges from a familiar, tangible context. The Amina mandazi scenario transfers the concept to a Kenyan everyday context, reinforcing KICD SLO (f): promoting use of real numbers in day-to-day activities.	Observable indicator: Each group's table is correctly completed with four rows of reciprocal values. Teacher checks tables during circulation — correct entries for 1÷(1/4)=4, 1÷(2/5)=5/2, 1÷(3/7)=7/3. Flag groups that invert incorrectly (e.g., write 1÷(2/5) = 2/5 instead of 5/2). The Amina mandazi solution (answer: 1 full batch, since 1÷(2/3)=3/2=1.5, so 1 full batch) reveals whether students can transfer the procedure to a word problem. Students who cannot set up the division from a word problem need scaffolding in Phase 3.

	division and connect to the fraction circle model.	number that fills the whole? Give me a word for that.' Guide toward the word 'reciprocal' if not surfaced. Write on the board: RECIPROCAL. Then say: 'Now read the Amina mandazi scenario and solve it the same way. 4 minutes.' Cold-call 1 group to present their solution. Provide WAIT TIME of 12 seconds before cold-calling after asking any question.		
Explain Phase 📹 **VIDEO:** [Finding the Reciprocal of Fractions] Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** [Identification and Writing of Reciprocal Fractions (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	The teacher formalises the definition: 'The reciprocal of a number n is 1 ÷ n, also written as 1/n. For a fraction a/b, the reciprocal is b/a. The product of a number and its reciprocal is always 1 — one whole pizza.' Students copy the definition and three worked examples into their books: (1) Reciprocal of 5 = 1/5 (verify: 5 × 1/5 = 1). (2) Reciprocal of 3/4 = 4/3 (verify: 3/4 × 4/3 = 1). (3) Reciprocal of √2 = 1/√2 = √2/2 (rationalised) — verify: √2 × (√2/2) = 2/2 = 1. Students then attempt 4 independent practice problems (mix of integers, fractions, and one irrational), checking their answer against the verification rule 'n × reciprocal(n) = 1.' Finally, a class discussion: 'Does every real number have a reciprocal? What about zero?' Students debate and arrive at the conclusion that zero has no reciprocal (division by zero is undefined).	Write on the board: 'RECIPROCAL RULE: n × (1/n) = 1.' Say verbatim: 'This is our key definition today. The reciprocal is the number that, when multiplied by the original, gives you exactly 1 — one whole pizza. Let us prove it works for three types of real numbers.' Work through Example 1 (integer), Example 2 (fraction), and Example 3 (irrational — √2) on the board, narrating each step aloud. For Example 3, say: 'Watch carefully — √2 is irrational. Its reciprocal is 1/√2. Mathematicians prefer no square root in the denominator, so we rationalise: multiply top and bottom by √2 to get √2/2. Now verify: √2 × √2/2 = 2/2 = 1. The rule still holds.' Ask: 'What surprises you about this?' Allow 10 seconds WAIT TIME. Cold-call 2 students. Then assign 4 independent practice problems. After 5 minutes, cold-call 4 different students (one per problem) to write answers on the board. Class checks each using the verification rule. Then pose the zero question: 'What is the reciprocal of zero? Discuss with your partner for 60 seconds.' Cold-call 2 pairs. Confirm: 'Zero has no reciprocal — you cannot divide 1 pizza into zero slices. This is why	Worked Example with Verification Loop: Each worked example is immediately followed by a multiplicative verification (n × 1/n = 1), creating a self-checking habit of mind. This is a deliberate sensemaking strategy — students learn that mathematics is internally consistent, and that they can verify their own answers without teacher confirmation. The extension to irrational numbers (√2) directly connects to Lessons 1–4 and advances the driving question: irrational numbers also have reciprocals, confirming they are fully-fledged members of the real number system.	Observable indicator: In independent practice, students correctly compute and verify all 4 reciprocals. Teacher reviews written work of 4–5 students during the practice window, focusing on (a) correct inversion of fractions, (b) correct rationalisation of irrational reciprocals, (c) correct multiplicative verification. The zero-reciprocal discussion reveals conceptual depth — students who say 'zero divided by one' instead of 'one divided by zero' reveal a persistent inversion misconception that must be addressed before Lesson 6.

		division by zero is undefined.'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Finding the Reciprocal of Fractions Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Identification and Writing of Reciprocal Fractions (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each student receives a sticky note. They write one new thing they now understand about real numbers AND one new question that today's lesson raised. Students post sticky notes on the class Driving Question Board under two columns: 'What We Now Know' and 'What We Still Wonder.' The class reviews 3–4 sticky notes together. The teacher explicitly connects the day's learning to the driving question: 'We now know that rational AND irrational numbers have reciprocals. Does this change your answer to: why can some numbers be written as exact fractions and others cannot — and how does knowing a number's type help us solve everyday problems in Kenya?'	Distribute sticky notes. Say: 'You have 2 minutes. On your sticky note, write: one thing you now understand that you did not understand at the start of today's lesson, AND one new question this lesson raised for you.' After 2 minutes, direct students to post notes on the DQB. Read 2 'What We Now Know' notes and 2 'What We Still Wonder' notes aloud. For each 'Wonder' note, ask the class: 'Can anyone partially answer this?' Allow 8 seconds WAIT TIME, then cold-call 1 student. Do not resolve all wonders — say: 'Some of these questions we will answer in Lessons 6 and 7, when we use mathematical tables and calculators to find reciprocals even faster.' Point to the driving question displayed on the board and say: 'Update your thinking: does knowing a number's reciprocal help you solve everyday problems in Kenya? Give me a thumbs up if yes, thumbs sideways if unsure, thumbs down if no.' Scan the room. Address any thumbs-down or sideways responses with a targeted prompt.	Driving Question Board (DQB) Sticky Note Protocol: The DQB is a visible, cumulative record of the class's growing understanding across the 7-lesson sequence. Posting both 'knows' and 'wonders' simultaneously validates both knowledge gain and intellectual curiosity. The teacher's explicit return to the driving question forces students to integrate today's evidence (reciprocals work for all real numbers) into their evolving explanation of the phenomenon.	Observable indicator: Every student posts a sticky note with both a 'know' and a 'wonder.' Teacher reads all notes after class to identify persistent misconceptions (e.g., 'reciprocals only work for fractions') and unresolved questions that signal readiness or unreadiness for Lesson 6. The thumbs survey provides a rapid whole-class comprehension check that is documented by the teacher for planning.
Model Building Phase 📹 **VIDEO:** Finding the Reciprocal of Fractions Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Identification and Writing of Reciprocal	Students return to their cumulative Real Numbers Model — a diagram they have been building since Lesson 1 showing the classification of real numbers. Today, they add a new layer: a 'Reciprocal' annotation next to each number type. Using a printed model revision template, students: (1) Write the reciprocal next to each example number already on their model (e.g., next to 3/4 they write 'reciprocal: 4/3'; next to $\sqrt{2}$ they write 'reciprocal: $\sqrt{2}/2$'). (2)	Distribute printed model revision templates (or direct students to their exercise book model page). Say: 'Open your Real Numbers Model. Today we are adding the reciprocal layer. You have 6 minutes to complete the four additions I am putting on the board.' Write the four tasks on the board. Circulate and check that: (a) reciprocal values are correct and verified, (b) the arrow annotation is present, (c) zero is correctly flagged, (d) the summary	Cumulative Model Revision: The Real Numbers Model is the students' personal, evolving scientific-mathematical model of the concept, revised after every lesson. Adding the reciprocal layer in direct connection to prior layers (number classification, rational vs. irrational) makes the learning cumulative and visible. The summary sentence anchoring the model to the pizza phenomenon reinforces the NGSS practice of Developing and Using Models and	Observable indicator: Model contains correct reciprocal annotations for at least 4 numbers (including one irrational), the directional arrow with label, a zero flag, and the summary sentence. Teacher collects 5–6 models at random after class for written feedback. Key errors to flag: missing rationalisation of irrational reciprocals (e.g., leaving $1/\sqrt{2}$ instead of $\sqrt{2}/2$), incorrect verification, or a summary sentence that does not mention

[Fractions (Flexbook)](#) Source: CK-12 🔍 Search ARES for similar readings	Draw an arrow labelled '÷ by this number = multiply by its flip' connecting the original number to its reciprocal. (3) Add a 'No reciprocal' flag next to zero. (4) Write one sentence at the bottom of their model: 'The reciprocal of a real number tells us how many of that number fit into 1 whole — just like knowing how many pizza slices fill a whole pizza.'	sentence connects back to the pizza phenomenon. After 6 minutes, say: 'Hold up your model. Partner, check: does your partner's model have all four additions?' Allow 1 minute of peer checking. Cold-call 2 students to read their summary sentence aloud. Affirm or gently correct. Close by saying: 'In Lesson 6, we will use mathematical tables to find reciprocals of numbers that are too complex to flip by hand — like 7-digit decimals. Your model will grow again.'	ensures the phenomenon remains the explanatory core, not a decorative detail.	the whole/pizza context, indicating the phenomenon connection has not been internalised.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) Did students successfully connect the procedural definition of reciprocal (1 ÷ n) to the conceptual meaning (how many slices fill the whole pizza)? If not, the fraction circle manipulative may need more time in Phase 2 of Lesson 6. (2) Which students struggled with rationalising irrational reciprocals (e.g., $1/\sqrt{2} \rightarrow \sqrt{2}/2$)? These students will need a short worked-example revisit at the start of Lesson 6 before moving to mathematical tables. (3) Did the Amina mandazi scenario generate genuine engagement, or did it feel forced? If students did not connect to it naturally, consider using a Kenyan fabric-measuring or recipe-scaling scenario in Lesson 6 as an alternative context. (4) Review the DQB sticky notes: how many 'wonders' are about zero, negative numbers, or very large numbers? These predict the misconceptions most likely to surface in Lesson 6 (reciprocals from tables and calculators). (5) How equitable was the cold-calling? Check your notes — were the same 3–4 students dominating responses? Use the name-card system more deliberately in Lesson 6 to ensure all students are called upon at least once across the lesson. (6) Core Competency check: did the group work in Phase 2 genuinely produce Communication and Collaboration, or did one student in each group dominate? Consider assigning explicit roles (Reader, Recorder, Calculator, Reporter) in the next lesson.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Students used fraction circle manipulatives to physically show how many equal slices fill a whole pizza, then recorded the relationship in a division table for slice sizes 1/3, 1/4, 2/5, and 3/7. They also solved the Amina mandazi scenario (1 kg flour ÷ 2/3 kg per batch) using reciprocal division.
What did I learn?	The reciprocal of a number n is 1 ÷ n (or b/a for a fraction a/b), and it represents how many of that quantity fit into 1 whole. The reciprocal exists for all real numbers — including irrationals like $\sqrt{2}$ (reciprocal = $\sqrt{2}/2$) — except zero, which has no reciprocal because division by zero is undefined.
How does this explain the phenomenon?	Knowing a number's reciprocal lets us solve everyday division problems in Kenya — such as sharing food equally, adjusting recipes, or measuring quantities in markets — by turning division into multiplication. This works whether the number is rational or irrational, confirming that all real numbers (except zero) behave consistently under this operation, which helps answer why knowing a number's 'type' matters in daily life.

LESSON 6 (40 minutes): Applying Reciprocals: Solving Real-World Computation Tasks	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Students apply reciprocals computed by three methods — manual division, mathematical tables, and calculators — to solve structured problems drawn from Kenyan everyday life, and justify which method is most appropriate for each situation.
Knowledge	Students know that the reciprocal of any non-zero real number n is 1 divided by n; that reciprocals of fractions a/b equal b/a; and that irrational numbers have reciprocals that are themselves irrational. Students also know when each computation method is most reliable.
Skills	Students can compute reciprocals using manual division, read reciprocal values from mathematical tables, and use a scientific calculator to find reciprocals of decimals, fractions, and irrational approximations. Students can apply these reciprocals to solve division problems in context and compare results across methods.
Attitudes	Students demonstrate accuracy, responsibility for tool use, and confidence in selecting the most appropriate method for a given real-life computation task.
Key Inquiry Question	How do we use real numbers in day-to-day activities?
Purpose in Storyline	This lesson is the culminating application lesson. Students have now learned what reciprocals are, how to find them by hand, from tables, and with a calculator. Here they integrate all three methods in Kenyan contexts — fertiliser distribution, market cost-splitting, and tile cutting — directly advancing the driving question by showing how knowing a number is rational or irrational changes how precisely we can compute and express its reciprocal.
Safety Notes	Calculators and devices should be handled with care. Students should not share devices in ways that cause crowding or distraction. No physical hazards are present in this lesson.

B. LESSON OVERVIEW

In a 40-minute lesson, students begin by predicting which computation method — manual division, tables, or calculator — they would choose for three Kenyan scenarios involving fertiliser bags in Eldoret, market cost-splitting at Gikomba, and tile-cutting dimensions in Mombasa. They then work in structured groups to solve all three problems using all three methods, recording results and discrepancies. In the Explain phase, groups share findings and the class debates method appropriateness. The Driving Question Board is updated with new evidence about rational versus irrational numbers in practical settings. Students close by sketching a decision-tree model showing when each reciprocal method is best. The anchoring pizza phenomenon is explicitly revisited to connect rational and irrational reciprocals back to slice measurement.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** [Finding the Reciprocal of]()	Students receive a one-page Problem Card with three scenario headers — (A) Fertiliser Distribution in Eldoret, (B) Market	Distribute the Problem Card face-down. Say: 'Do not flip yet. Listen carefully.' Read each scenario title aloud with one sentence of	Prediction with justification activates prior knowledge from Lessons 3, 4, and 5. The three-method framework provides	Listen for whether students connect method choice to number type — for example, whether they say tables are less useful for

Fractions Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Identification and Writing of Reciprocal Fractions (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Cost-Splitting at Gikomba, and (C) Tile-Cutting in Mombasa — and three method options labelled: Method 1: Manual Division, Method 2: Mathematical Tables, Method 3: Calculator. Without solving anything yet, each student individually writes in their notebook which method they predict is best for each scenario and gives one reason. They then share their prediction with a shoulder partner for 90 seconds before the teacher cold-calls three pairs to share aloud. Teacher writes the three most common predictions on the board under the heading 'Our Predictions' for later comparison.	context. Then say: 'Flip the card. Look only at the method choices. Predict — do not calculate — which method you would trust most for each scenario. You have two minutes. Work alone first.' Set a visible 2-minute timer. After individual thinking, say: 'Now share your prediction and your reason with your partner. One minute.' After pair-sharing, cold-call three pairs using name sticks. For each response, ask: 'What evidence from our earlier lessons made you choose that method?' Apply a WAIT TIME of 8 seconds after each cold-call before accepting or redirecting. Record predictions on the board without evaluating them. Say: 'We will return to these predictions after we actually solve the problems.'	cognitive scaffolding, and the peer-share step surfaces diverse reasoning before group work begins.	irrational numbers because the table only gives decimal approximations. Note students who conflate accuracy with speed; this will guide teacher questioning during the Observe phase.
Observe Phase 📹 **VIDEO:** Finding the Reciprocal of Fractions Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Identification and Writing of Reciprocal Fractions (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Groups of three (one per scenario as lead recorder) open the Problem Card fully. The three problems are: (A) A farmer in Eldoret has 1 bag of fertiliser and wants to distribute it equally among 7 plots. What fraction of the bag does each plot receive? Compute this reciprocal by all three methods and record results. (B) Four traders at Gikomba split a market stall cost of Ksh 1 equally. What does each trader pay as a fraction of the total? Now the stall owner changes the number of traders to a number whose reciprocal is irrational — she proposes dividing by the square root of 4. Discuss: is that reciprocal rational or irrational? Compute it all three ways. (C) A tile-cutter in Mombasa needs to cut a 1-metre tile into pieces each of length 1 divided by the square root of 2 metres. Compute the	Circulate systematically — spend approximately 3 minutes with each of four groups. At each group, ask a targeted question rather than providing answers. Suggested prompts: 'How are you finding the reciprocal of 7 by hand — show me the division setup.' 'When you read the table for 7, which column did you look at and why?' 'What does your calculator show for 1 divided by root 2 — how does that compare with the table value?' 'For scenario B, someone said root 4 is actually rational — do you agree? How would that change things?' Apply a WAIT TIME of 10 seconds after each probe before following up. If a group finishes early, pose an extension: 'What is the reciprocal of the reciprocal of 7? Does that pattern hold for irrational numbers too?' Flag two or three observations to highlight in the Explain phase.	The structured results table enforces comparison across methods and makes discrepancies visible. Scenario B is deliberately designed to surface the insight that root 4 equals 2, which is rational — connecting back to number classification from Lesson 2. Scenario C forces engagement with irrational reciprocals in a physical context, echoing the pizza arc-length situation from the anchoring phenomenon.	Check that students are recording method differences accurately, not just copying the calculator answer three times. Ask any group that shows identical entries for all three methods: 'Did you actually use the table, or did you copy the calculator value?' Look for whether students can articulate why the table gives only an approximation for irrational numbers. Collect mental notes on which groups correctly identify root 4 as rational.

	reciprocal of root 2 to four decimal places using each method, then state how many whole pieces fit and what the leftover length is. Students fill in a structured results table with columns: Scenario, Method 1 result, Method 2 result, Method 3 result, Difference between methods, Most precise method chosen. Groups have 15 minutes for this phase.			
Explain Phase 📹 **VIDEO:** Finding the Reciprocal of Fractions Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Identification and Writing of Reciprocal Fractions (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each group sends one spokesperson to present their results table for one scenario — teacher assigns which scenario each group presents to ensure all three are covered. Spokesperson reads the results aloud and states: which method gave the most precise answer, whether the number involved was rational or irrational, and which method they would recommend in a real Kenyan setting and why. After all three presentations (approximately 2 minutes each), the teacher leads a whole-class comparison using a board table that aggregates all results. Class discusses: (1) Why do manual division and tables sometimes give slightly different answers from the calculator? (2) For which scenario was the number rational and for which irrational — and how could you tell from the decimal expansion? (3) In a rural area of Kenya with no electricity, which method is safest to rely on? Students update their individual notebooks with a written summary sentence: 'For rational numbers, all three methods give the same exact value because... For irrational numbers, methods differ because...'	After each group presentation, ask the class: 'Does any other group have a different result for that scenario? If so, stand up.' If discrepancies exist, say: 'Let us trace where the difference came from — was it in the table reading, the division, or the rounding on the calculator?' Apply a WAIT TIME of 8 seconds. For the whole-class board comparison, use deliberate questioning: 'Look at scenario C — the tile problem. The calculator shows 0.7071. The table may show 0.7070 or 0.7072. Why does this difference matter for a tile-cutter in Mombasa — and when does it NOT matter?' After responses, connect explicitly to the phenomenon: 'Remember our pizza diagram from Lesson 1. The slice that was irrational — its size could never be written as an exact fraction. That is exactly what we see in scenario C. The tile length is irrational, so no method gives us a perfect exact decimal — they all give approximations. That is a property of irrational numbers, not a mistake.' Pause for reflection. Say: 'Write your summary sentence now. Two minutes of silent writing.'	Cross-group comparison exposes productive discrepancies that make the limitation of each method concrete. The teacher explicitly bridges back to the anchoring phenomenon to ensure the driving question is advanced — irrational numbers produce non-terminating, non-repeating decimals that no tool can capture exactly, only approximate. The summary sentence writing consolidates the conceptual distinction.	Listen to summary sentences during a brief share-out. Correct any student who writes 'the calculator is always more accurate' — prompt them to consider that calculators also round irrational values. Check that students can identify scenario B's twist (root 4 = 2 is rational) and that they articulate this distinction clearly.
Driving	Students return to the class	Say: 'Look at our board. We have	The two-colour sticky note system	Review posted green notes after

Question Board (DQB) Creation 📹 **VIDEO:** [Finding the Reciprocal of Fractions] Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** [Identification and Writing of Reciprocal Fractions (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Driving Question Board. The teacher reads aloud the current posted questions from previous lessons that relate to today. Students are given two sticky notes each. On the first (green), they write one question from the board that today's lesson has now answered. On the second (yellow), they write one new question or remaining uncertainty that today's work raised for them. Students post both notes on the DQB. Teacher reads a selection of green notes aloud to affirm progress, then reads two or three yellow notes to acknowledge remaining curiosity. The class briefly votes by show of hands on which yellow question they most want answered in Lesson 7.	been building evidence for six lessons now. Today we applied reciprocals in three real Kenyan situations. Which question from this board did today answer for you? Take one minute and write it on your green note — be specific, not just the question number.' After posting, select three green notes and read them aloud: 'This group says today answered the question about whether irrational numbers have reciprocals — what evidence from today supported that?' WAIT TIME of 6 seconds. Then say: 'Your yellow note is equally important. What does today make you wonder that we have NOT yet answered?' After posting yellow notes, point to the two or three most common themes and say: 'In Lesson 7 we will build our Real Number Map. These questions will guide what you put in it. Hold onto them.'	separates resolved questions from emerging ones, making the cumulative progress of the inquiry visible. Voting on the priority yellow question gives students agency in shaping the final lesson and reinforces that scientific and mathematical inquiry generates new questions even as it answers old ones.	class. If most students mark the same question as answered, that concept is secure. If many yellow notes show the same confusion — for example, 'can a reciprocal ever be irrational and rational at the same time?' — plan to address this explicitly at the start of Lesson 7.
Model Building Phase 📹 **VIDEO:** [Finding the Reciprocal of Fractions] Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** [Identification and Writing of Reciprocal Fractions (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	In the final 6 minutes, students individually sketch a simple decision-tree model in their notebooks titled 'Which Reciprocal Method Should I Use?' The tree has three decision nodes: (1) Is the number rational or irrational? (2) Do I have access to electricity and a calculator? (3) Do I need more than 4 decimal places of accuracy? Each path through the tree leads to a recommendation box: Manual Division, Mathematical Tables, or Calculator. Students annotate each recommendation box with one Kenyan example from today's lesson that fits that path. This model will be incorporated into the full Real Number Map in Lesson 7.	Say: 'In Lesson 7 you will build your complete Real Number Map. Today, as a foundation, you will sketch a decision tree — a tool that shows thinking steps. I will draw the three nodes on the board. Your job is to fill in the paths and the examples from today.' Draw the three nodes quickly on the board as labelled empty diamonds. Say: 'You have five minutes. Use examples from our three scenarios — Eldoret, Gikomba, Mombasa.' Circulate and check that students are writing Kenyan examples rather than generic ones. For students who finish early, prompt: 'Can you add a fourth node — what if the number is very large, like a 6-digit fertiliser distribution number? Which method would you choose	The decision-tree model externalises conditional reasoning, a key mathematical thinking skill. Anchoring the nodes in real Kenyan decisions — electricity access, precision requirements — makes the model culturally relevant and practically useful. The model also previews the hierarchical classification structure that will anchor Lesson 7.	Collect or photograph two to three student decision trees at the end of class. Check that paths are logically consistent — for example, that students do not recommend tables for irrational numbers requiring 6-digit accuracy. Use these as discussion anchors at the start of Lesson 7.

		then?' In the final minute, ask two students to read their Eldoret path recommendation aloud. Affirm accurate responses and gently redirect errors by saying: 'What does the table give us for a 6-digit number — is there a table entry for that?' WAIT TIME of 5 seconds.		

D. TEACHER REFLECTION
After the lesson, consider the following: Did students successfully distinguish between rational and irrational reciprocals when the context changed — as in scenario B where root 4 is rational? Were students able to articulate why method differences arise for irrational numbers without simply saying the calculator is always right? Did the three Kenyan contexts feel meaningful to students, or did any group disengage from one scenario? Were the 15 minutes of group work enough time for all three scenarios, or did some groups only complete two — and if so, which scenario should be shortened or scaffolded further? Review the yellow sticky notes from the DQB to ensure Lesson 7 addresses the most pressing remaining questions. Consider whether any student showed a persistent misconception — for example, believing that a longer decimal is always more accurate — and plan a targeted correction at the start of Lesson 7.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Students observed that computing the reciprocal of a rational number such as 1 divided by 7 gives consistent results across all three methods, while computing the reciprocal of an irrational number such as 1 divided by root 2 gives slightly different approximations depending on the method used. They also observed that root 4, though written with a radical sign, equals 2 and is rational — its reciprocal is exact.
What did I learn?	Students learned that all three reciprocal methods are valid but serve different purposes: manual division is always available and exact for rational numbers; mathematical tables are reliable and portable for up to 4-digit rational values; calculators provide fast approximations including for irrational numbers but always round the result. They learned that irrational reciprocals cannot be expressed exactly by any method.
How does this explain the phenomenon?	Students explained that the choice of reciprocal method depends on three factors — the type of number (rational or irrational), the tools available (calculator, tables, or neither), and the precision required for the real-world task. They connected this back to the pizza phenomenon by explaining that a slice whose size is irrational — like an arc length involving root 2 — can never be expressed as an exact fraction or terminating decimal, only approximated, just as its reciprocal can only be approximated.

LESSON 7 (80 minutes): Final Explanation: Putting It All Together — Real Numbers, Reciprocals, and Everyday Life in Kenya	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To consolidate and synthesise all learning from the unit by constructing a final explanation of real numbers and reciprocals, comparing initial and final mental models, completing the Driving Question Board, and applying knowledge to authentic Kenyan everyday contexts.
Knowledge	Learners demonstrate that: (a) whole numbers are classified as odd, even, prime, and composite; (b) real numbers are classified as rational or irrational; (c) reciprocals of real numbers are determined by division, mathematical tables, and calculators; (d) reciprocals are applied in mathematical computations relevant to daily life.
Skills	Learners are able to: classify real numbers accurately; calculate reciprocals using multiple methods (division, tables, calculator); explain why some numbers cannot be expressed as exact fractions; apply reciprocals to solve real-world problems; construct and critique scientific explanations using evidence gathered across the unit.
Attitudes	Learners promote the use of real numbers in day-to-day activities (KICD SLO f); take responsibility for accurate mathematical reasoning; demonstrate unity by co-constructing the final class explanation; develop self-esteem by articulating their own growing understanding (PCI: Life Skills).
Key Inquiry Question	How do we use real numbers in day-to-day activities?
Purpose in Storyline	This is the capstone lesson of the unit. Students have spent six lessons gathering evidence: classifying numbers, exploring rational and irrational decimals, computing reciprocals by division, reading mathematical tables, and using calculators. In this lesson, they return to the anchoring phenomenon — the Unequal Pizza Slices Problem — with full explanatory power. They compare their Lesson 1 model (naïve) with their final model (evidence-based), complete the Driving Question Board, and write a Final Explanation that answers the driving question using all accumulated evidence.
Safety Notes	No physical safety concerns. Remind learners to handle calculators and mathematical tables with care (KICD Value: Responsibility). Ensure inclusive participation — no learner should be made to feel their Lesson 1 model was 'wrong'; frame growth as the goal.

B. LESSON OVERVIEW
In this final lesson of Sub-Strand 1.1, learners synthesise seven lessons of evidence to construct a complete, class-level Final Explanation addressing the driving question: 'Why can some numbers be written as exact fractions and others cannot — and how does knowing a number's type help us solve everyday problems in Kenya?' The lesson opens by returning to the anchoring phenomenon (the Unequal Pizza Slices Problem) and asking learners to articulate what they now know that they did not know in Lesson 1. Learners then compare their original (Lesson 1) models with their revised models, identifying the conceptual journey. In small groups they draft a Final Explanation using a structured evidence-based writing frame, drawing on reciprocal computation (division, tables, calculator), classification of real numbers, and Kenyan contexts (sharing ugali, splitting land in the Rift Valley, calculating rates for boda-boda fares). The class converges on a shared Final Explanation, closes out all remaining DQB questions, and each learner completes a personal reflection on their learning growth. The lesson closes with a transfer task set in a new Kenyan context to confirm readiness for Sub-Strand 1.2.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** Prime numbers Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Real Number Line Graphs (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners receive their original Lesson 1 'prediction strip' (or recall it from their notebooks) where they first described — in their own words — why some numbers seem 'neat' and others do not. Without referring to any notes, each learner spends 3 minutes writing a 'brain dump': everything they now know about real numbers, reciprocals, rational and irrational numbers. They then self-compare: What did I think in Lesson 1? What do I know now? What changed? Learners mark their own growth using a simple 3-column table: 'I used to think… / I now know… / The evidence that changed my mind was…'	Display the anchoring phenomenon image (two pizza diagrams: rational slices vs. irrational arc-length slice) on the board — the same image from Lesson 1. Say verbatim: 'We started this unit staring at these two pizza slices. You had questions. You had guesses. Before we do anything else today, I want you to show yourself — not me — how much your thinking has grown. Open to a fresh page. Three minutes. Brain dump. Everything you know. Go.' Set a visible 3-minute timer. Maintain silence — do not prompt. After 3 minutes say: 'Now look at your very first entry in this unit. In your table, write what you used to think, what you now know, and the single piece of evidence that most changed your mind. Two minutes.' Cold-call 3 learners to share one row of their table. After each, ask the class: 'Does anyone want to add to that? Thumbs up if that evidence also changed your thinking.' Wait 10 seconds after each cold-call before accepting additions.	Strategy: 'I Used to Think / Now I Know' (Metacognitive Reflection). This strategy makes thinking visible and celebrates conceptual growth. It connects every learner's current understanding back to the anchoring phenomenon, reinforcing that the unit was one long investigation into the puzzle of the pizza slices. It also activates prior knowledge from all six preceding lessons before new synthesis begins.	Observable indicator: Each learner's 3-column table contains at least one accurate mathematical claim in the 'I now know' column (e.g., a correct definition of rational/irrational, or a correct statement about reciprocals). Teacher circulates and reads 5–6 tables during the 2-minute writing period, noting any persistent misconceptions for targeted questioning in Phase 3. Flag any learner who still conflates 'irrational' with 'impossible' for a check-in during group work.
Observe Phase 📹 **VIDEO:** Prime numbers Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Real Number Line Graphs (Flexbook) Source: CK-12	Learners retrieve the cumulative model they have been building and revising across Lessons 1–6 (a labelled diagram showing the real number system, with examples of rational and irrational numbers, a reciprocal notation, and at least one Kenyan context). They place their Lesson 1 model (a simple sketch, likely unlabelled or incorrectly labelled) side by side with their current model. In pairs, they conduct a 'Model Comparison Protocol': (1) Name one thing that	Distribute or ask learners to open their model pages. Say: 'Put your Lesson 1 sketch on the left. Put today's model on the right. You have 4 minutes with your partner to complete the Model Comparison Protocol on the board.' Post the three protocol questions visibly. After pair work, say: 'We are doing a 4-minute gallery walk. You will read two other pairs' comparisons. Leave a sticky note with one star — something they noticed that you also noticed — and one arrow —	Strategy: Model Comparison Protocol with Gallery Walk. Making the evolution of the model visible helps learners see that knowledge is constructed iteratively from evidence — a core scientific and mathematical practice (NGSS SEP 2: Developing and Using Models). The gallery walk creates a low-stakes peer-review environment that reinforces mathematical communication and collaboration (KICD Core Competency).	Observable indicator: Each pair can name at least one specific lesson/activity as the source of a model revision (e.g., 'In Lesson 5 when we used the calculator and found that √2 never repeats, we changed our model to show irrational numbers are non-terminating and non-repeating'). Teacher notes whether learners are connecting evidence to model changes — if pairs struggle, prompt: 'What did we discover when we tried to write 1 ÷ 3 on the

🔍 Search ARES for similar readings	is in the final model but missing from the L1 model. (2) Name one thing in the L1 model that was incorrect — and explain what the correct understanding is. (3) Name the specific lesson or activity that caused that revision. Pairs then do a gallery walk — 4 minutes — reading two other pairs' model comparisons posted on the wall.	something they could add.' After the gallery walk, cold-call 2 pairs to share the most surprising difference between their L1 and final model. Ask: 'What does that difference tell us about how science — and mathematics — actually works?' Wait 15 seconds. Accept 2–3 responses. Validate: 'Models in mathematics, just like in science, are always works in progress. Today we finalise ours — for now.'		calculator? How did that change your diagram?'
Explain Phase 📹 **VIDEO:** Prime numbers Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Real Number Line Graphs (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	In groups of 4, learners construct a written Final Explanation using a structured evidence frame. The frame has four sections: (A) Claim — a direct answer to the driving question in 1–2 sentences; (B) Evidence — at least 3 pieces of mathematical evidence from across the unit (must include at least one from tables, one from calculator work, and one from the pizza/fraction circle phenomenon); (C) Reasoning — explaining why the evidence supports the claim, using correct mathematical vocabulary (rational, irrational, reciprocal, non-terminating, non-repeating); (D) Kenyan Connection — one real-world example from Kenya showing how knowing a number's 'type' helps solve a practical problem. Groups write their Final Explanation on A3 paper. Each group must include: a worked reciprocal example (by division OR tables OR calculator), a correct classification of at least 4 numbers (e.g., √3, 0.75, π, 7), and one Kenyan context sentence (e.g., 'A boda-boda rider in Kisumu charging Ksh 150 per km needs the reciprocal of 150 to find the distance per shilling').	Distribute A3 paper and the evidence frame template. Say: 'Your group's job is to answer the driving question — for real, with evidence — as if you were explaining it to a Form 1 student who has never heard of irrational numbers. Use the frame. Every section must be filled. You have 15 minutes.' Set a 15-minute timer. Circulate with targeted questions: To groups struggling with the Claim: 'Start with the pizza. Can you always write a slice as a fraction? What does that depend on?' To groups with weak Reasoning: 'You said √2 is irrational. Why does that mean its reciprocal behaves differently? What did the tables show us?' To strong groups: 'Can you add a second Kenyan context — maybe from farming in the Rift Valley or sharing githeri equally among different numbers of people?' At 12 minutes, give a 3-minute warning. After 15 minutes, cold-call each group to read their Claim sentence only. Ask the class to vote (thumbs up / sideways / down) on each claim's accuracy. For any thumbs-sideways, ask: 'What one word would make this claim more precise?' Wait 10 seconds before	Strategy: Claim-Evidence-Reasoning (CER) Writing Frame with Kenyan Connection. CER is an NGSS-aligned sensemaking strategy (SEP 6: Constructing Explanations) that scaffolds scientific and mathematical argumentation. Adding the 'Kenyan Connection' section ensures the explanation is not abstract — it roots mathematical understanding in the learners' lived reality (Nairobi, Kisumu, Rift Valley), fulfilling KICD SLO (f): promote the use of real numbers in day-to-day activities.	Observable indicators: (1) The Claim directly references both rational AND irrational numbers and the concept of exact fractions. (2) At least one evidence item is a specific numerical example (e.g., '1/3 = 0.3333… from the tables, which never terminates'). (3) The Kenyan Connection sentence contains a real, computable scenario — not just a vague mention of 'daily life'. Teacher uses a 3-point rubric (Developing / Meets / Exceeds) on a clipboard to rate each group's Claim and Evidence before the share-out, using this to select the order of cold-calls (Developing groups called first so feedback can inform the others).

		accepting responses.		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Prime numbers Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Real Number Line Graphs (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	The full class gathers at the Driving Question Board (DQB), which has been built across all 7 lessons. Learners read all remaining sticky notes that are still in the 'unanswered' column. In a class discussion, they determine: (A) Which questions can now be answered using what they know — and they physically move those cards to the 'answered' column, writing the answer on the back; (B) Which questions are partially answered — they annotate with what they know; (C) Which questions point toward future learning (Sub-Strand 1.2: Indices and Logarithms) — they place these in a 'Coming Next' column. Each learner then adds one final sticky note to the DQB: 'The biggest idea I am taking away from this unit is…' These are read aloud by volunteers.	Stand at the DQB. Say: 'Seven lessons ago, you stuck questions on this board because the pizza problem puzzled you. Let's see how many we can close today.' Read each unanswered question aloud. For each, ask: 'Who can answer this — using evidence from our unit? Give me a full sentence.' Wait 15 seconds before cold-calling. For each answer, ask: 'Do we agree? Thumbs up if this question is now fully answered.' Physically move cards with the learner who answered. For questions that are partially answered, say: 'We have part of the answer. What part do we know? What part is still open?' For questions that point to Sub-Strand 1.2 (e.g., 'What happens when you raise an irrational number to a power?'), say: 'Great — this is exactly where we are heading next. File it in Coming Next.' End by saying: 'Now, one sticky note each. The biggest idea you are taking away. Stick it on the board. These are yours — and they are evidence of real thinking.' Give 2 minutes for writing, then ask 4–5 volunteers to read theirs aloud. Celebrate the board: 'Look at what you built. This is a thinking record of 7 lessons of mathematics.'	Strategy: DQB Final Closure with 'Answered / Partial / Coming Next' Sort. This strategy (aligned with NGSS SEP 1: Asking Questions and SEP 8: Obtaining, Evaluating and Communicating Information) makes the entire unit's inquiry arc visible and celebrates the closure of genuine student questions. The 'Coming Next' column creates a motivating bridge to Sub-Strand 1.2 by showing learners that their curiosity is generative — their questions drive the curriculum forward.	Observable indicator: At least 80% of DQB questions that were generated in Lessons 1–3 are moved to 'answered' with a complete, evidence-based sentence on the back. Teacher reads the 'biggest idea' sticky notes after class to identify any remaining conceptual gaps (e.g., if multiple learners write 'I learned that irrational numbers are very big' — a misconception — this is flagged for a brief correction at the start of Sub-Strand 1.2 Lesson 1).
Model Building Phase 📹 **VIDEO:** Prime numbers Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos	Each learner makes final revisions to their personal cumulative model of the real number system. They must ensure it includes: (1) a correct classification hierarchy (Real Numbers → Rational / Irrational; Rational → Integers → Whole → Natural; Irrational examples: $\sqrt{2}$, π, $\sqrt{3}$); (2) at least one annotated reciprocal example	Say: 'Before you finalise your model, ask yourself: if I showed this to a Form 1 student, could they understand the whole real number system from it? If not, add what is missing.' Give 5 minutes for final model revision. Then distribute the Transfer Task. Say: 'This is individual. It is a new context — Amina's farm — but you	Strategy: Final Model Revision + Transfer Task (NGSS SEP 2 and SEP 6). The transfer task places the unit's core ideas in a brand-new Kenyan agricultural context (Amina's farm in the Rift Valley), requiring learners to apply — not recall — their understanding. This distinguishes surface learning from deep, transferable competency,	Observable indicators for Transfer Task: (a) Correct classification of $\sqrt{5}$ as irrational (non-terminating, non-repeating decimal) and 3/4 as rational (exact fraction / terminating decimal). (b) Correct setup of division as $(3/4) \div \sqrt{5}$ and conversion to multiplication using reciprocal: $(3/4) \times (1/\sqrt{5})$. (c) Correct use of tables or calculator

📖 **HTML:** [Real Number Line Graphs (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	showing all three methods — by division, by tables, by calculator; (3) a 'Kenyan context' label — a real situation where this knowledge applies. Learners then complete a Transfer Task individually (10 minutes): 'Amina is a maize farmer in the Rift Valley. She has a rectangular plot whose width is √5 metres and whose length is 3/4 km. (a) Classify each measurement as rational or irrational and explain why. (b) Amina wants to find how many widths fit into the length. Write this as a division, then use the reciprocal to solve it. Show your method using mathematical tables OR a calculator. (c) What does your answer tell Amina practically?' Learners submit this as the unit's Exit Assessment.	have every tool you need. Show all working. You have 10 minutes.' Circulate silently. At 5 minutes, say: 'Check: have you classified both measurements? Have you shown the reciprocal step explicitly?' Do not answer content questions — redirect: 'What does your model show you?' After 10 minutes, collect Transfer Tasks. Close with: 'The pizza puzzle that opened this unit — can you answer it now? Turn to your neighbour. 30 seconds. Answer the driving question in one sentence.' Cold-call 2 pairs. Say verbatim: 'You walked in seven lessons ago wondering why some slices could be written as neat fractions and others could not. Today you can explain it — with evidence, with mathematics, and with examples from your own country. That is what real mathematical thinking looks like.'	fulfilling KICD Core Competency: Critical Thinking and Problem Solving. The final model revision ensures the visual, cumulative artefact of the unit is complete and accurate.	to evaluate 1/√5 ≈ 0.4472. (d) A practical interpretation sentence (e.g., 'Approximately 0.335 widths fit into the length, meaning the width is much smaller than the length'). Full marks require all four. Partial credit for correct classification + correct reciprocal setup even if computation contains arithmetic errors.

D. TEACHER REFLECTION
After this final lesson, reflect on the following questions before beginning Sub-Strand 1.2: (1) UNIT ARC: Did the anchoring phenomenon (Unequal Pizza Slices) sustain curiosity across all 7 lessons, or did it feel forced by Lesson 5–6? What would you change about the phenomenon for next year? (2) MODEL GROWTH: When you compared learners' Lesson 1 and final models, what was the most common conceptual shift? Was there a persistent misconception (e.g., 'irrational means impossible' or 'reciprocal means opposite') that survived all 7 lessons? Plan a 5-minute correction for the start of Sub-Strand 1.2. (3) DQB QUESTIONS: Which DQB questions were learners unable to answer by Lesson 7? Do any of these belong in Sub-Strand 1.2 (Indices and Logarithms)? Flag these for your Lesson 1 DQB seeding activity in the next unit. (4) TRANSFER TASK: What percentage of learners correctly set up the reciprocal step in Amina's farm problem? If fewer than 70% succeeded, consider a 10-minute reciprocal fluency warm-up in Sub-Strand 1.2 Lesson 1. (5) EQUITY CHECK: Did all learners — including those who were hesitant in Lesson 1 — feel safe to share a 'biggest idea' on the DQB? The 'I Used to Think / Now I Know' frame is designed to validate growth, not just accuracy. Note any learners who still appeared disengaged and plan a personal check-in. (6) KICD VALUES: Did the lesson authentically promote Unity (collaborative Final Explanation) and Responsibility (care of mathematical tables and calculators)? Were the Kenyan contexts (boda-boda, Rift Valley farming, githeri sharing) genuinely motivating, or did they feel tokenistic? Adjust for local relevance based on your specific school community.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners observed (across the full unit, reviewed in this lesson): two pizza diagrams — one divided into rational-fraction slices (1/4, 1/3, 1/8) and one where a slice is expressed as an irrational arc length (≈ 0.7071…). They observed fraction circles as

	manipulatives showing that some divisions produce terminating decimals and others do not. They observed mathematical table outputs for reciprocals of both rational and irrational numbers, and calculator displays showing non-terminating, non-repeating decimals for √2, √3, and π.
What did I learn?	Learners learned that: real numbers are classified as rational (expressible as p/q, where q ≠ 0) or irrational (non-terminating, non-repeating decimals such as √2, √3, π); the reciprocal of a number n is 1/n, found by division, mathematical tables, or calculator; reciprocals are used to solve division problems (e.g., how many slices fit into a whole); knowing a number's type (rational vs. irrational) predicts whether its reciprocal can be written as a neat fraction; whole numbers are further classified as odd, even, prime, and composite (underpinning the rational/irrational distinction).
How does this explain the phenomenon?	Learners explained that some pizza slice sizes can be written as exact fractions (rational numbers) because they result from dividing a whole into a countable, finite number of equal parts expressible as p/q. Other slice sizes — such as those involving π or √2 — are irrational: their decimal expansions never terminate and never repeat, so no exact fraction can capture them. The reciprocal tells us how many of one quantity fit into another (e.g., how many slices fill the whole pizza), and this works for both rational and irrational numbers — but the reciprocal of an irrational number is itself irrational. Knowing a number's type helps Kenyans solve real problems: sharing land equally in the Rift Valley, calculating boda-boda fares per shilling in Kisumu, or dividing githeri portions fairly — because it tells you whether an exact fractional answer exists or whether an approximation is the best you can do.

DIFFERENTIATION AND INCLUSION		
Learner Need	**Support Strategies**	**Extension Strategies**
English Language Learners	Provide bilingual glossaries; allow use of first language for initial model drawing.	Write extended explanations of observations; lead group discussions in English.
Students with learning difficulties	Provide structured note frames; pair with a supportive partner during investigations.	Mentor peers; create additional visual models to explain concepts.
Gifted and advanced learners	Access to additional reading materials; challenge questions during DQB.	Lead model-building discussions; research and present extension topics to class.